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Cheng et al.

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(54) **SEMICONDUCTOR DEVICES AND METHODS OF FABRICATING THE SAME**

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None
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,214,719 B1 * 4/2001 Nag H01L 21/7682
438/622
2015/0255578 A1 * 9/2015 Chang H01L 29/7848
438/300

(Continued)

FOREIGN PATENT DOCUMENTS

TW 1737136 B 8/2021
TW 202209415 A 3/2022
TW 202209564 A 3/2022

OTHER PUBLICATIONS

Cheng, Jung-Chien et al "Hybrid Semiconductor Device" U.S. Appl. No. 17/226,851; Assignee: Taiwan Semiconductor Manufacturing Company, Ltd. 38 pages of Specification, 37 pages of Drawings.

(Continued)

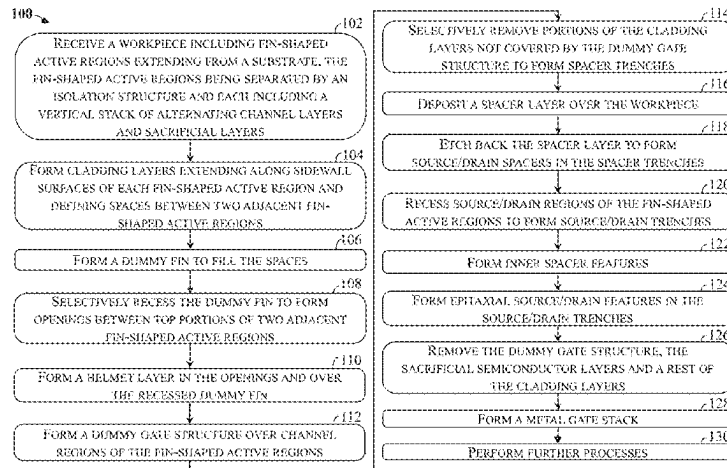
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(57) **ABSTRACT**

Semiconductor devices and methods are provided. An exemplary method according to the present disclosure includes providing fin-shaped active regions protruding from a substrate, forming cladding layers extending along sidewalls of the fin-shaped active regions, forming a dielectric feature over the substrate to fill space between two adjacent cladding layers, forming a gate structure over channel regions of the fin-shaped active regions and over a first portion of the cladding layers, performing an etching process to remove a second portion of the cladding layers not covered by the gate structure to form sidewall spacer trenches, forming a dielectric spacer in each of the sidewall spacer trenches, and after the forming of the dielectric spacers, forming source/drain features.

20 Claims, 25 Drawing Sheets



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H10D 62/10 (2025.01)
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 (2025.01); *H10D 30/6757* (2025.01); *H10D*
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(56) **References Cited**

U.S. PATENT DOCUMENTS

2020/0075423 A1 * 3/2020 Kwok H01L 21/823481
 2020/0343376 A1 * 10/2020 Chiang H01L 29/66439
 2021/0098302 A1 * 4/2021 Ju H10B 10/00
 2021/0343645 A1 11/2021 Peng
 2021/0375857 A1 12/2021 Huang

OTHER PUBLICATIONS

Chiang, Kuo-Cheng et al "Nanosheet Devices With Hybrid Structures And Methods Of Fabricating The Same" U.S. Appl. No. 17/520,385, Assignee: Taiwan Semiconductor Manufacturing Company, Ltd. 35 pages of Specification, 57 Pages of Drawings.

* cited by examiner

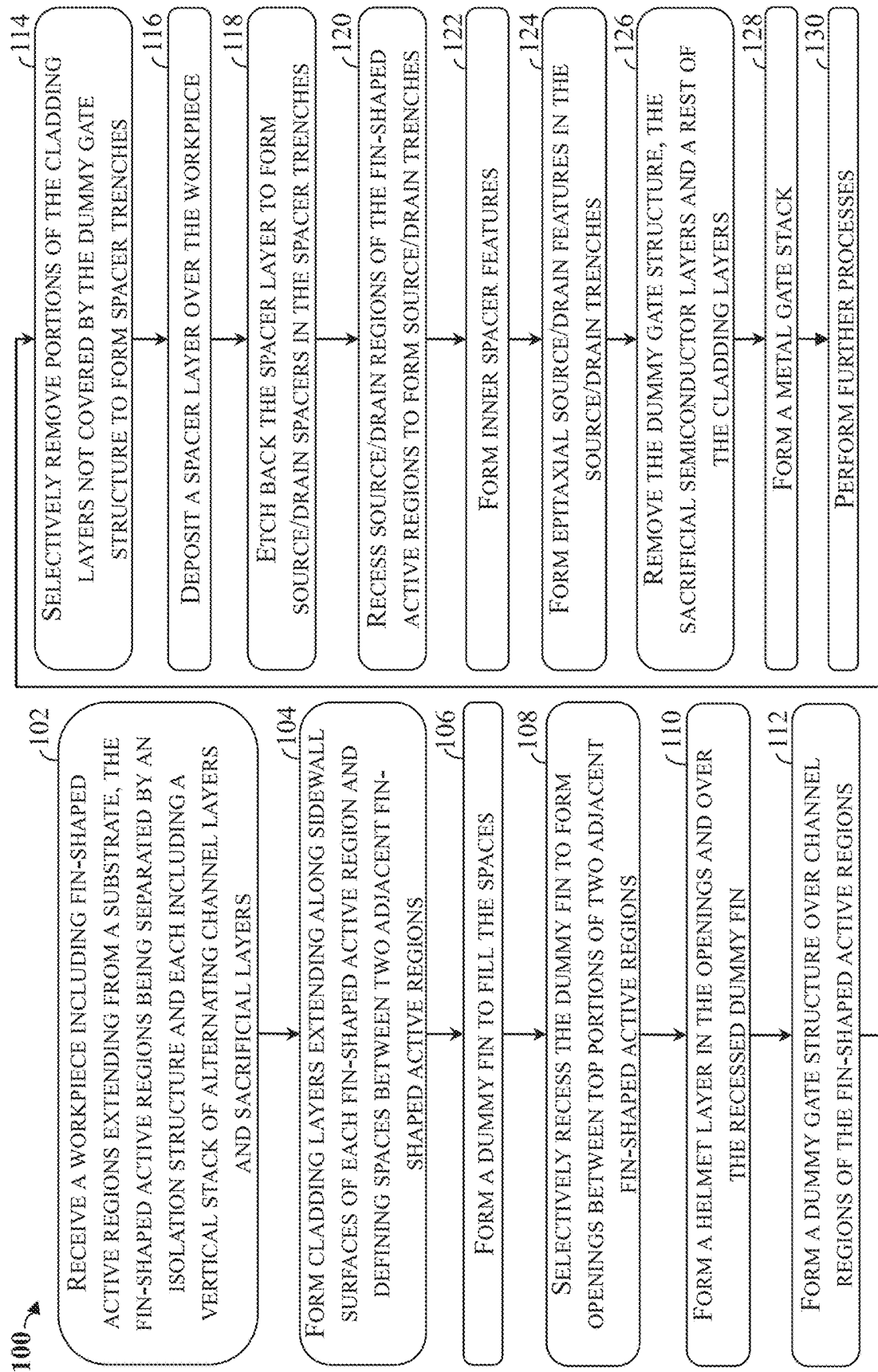
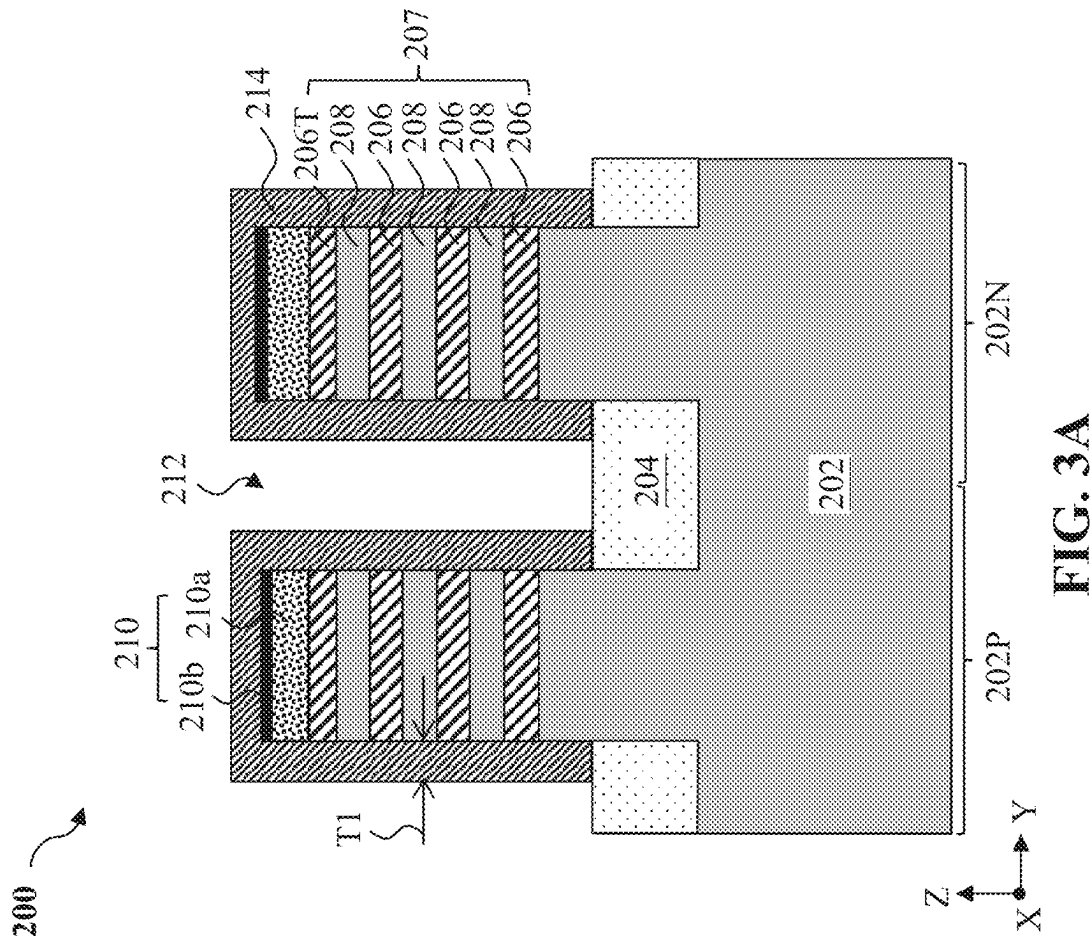
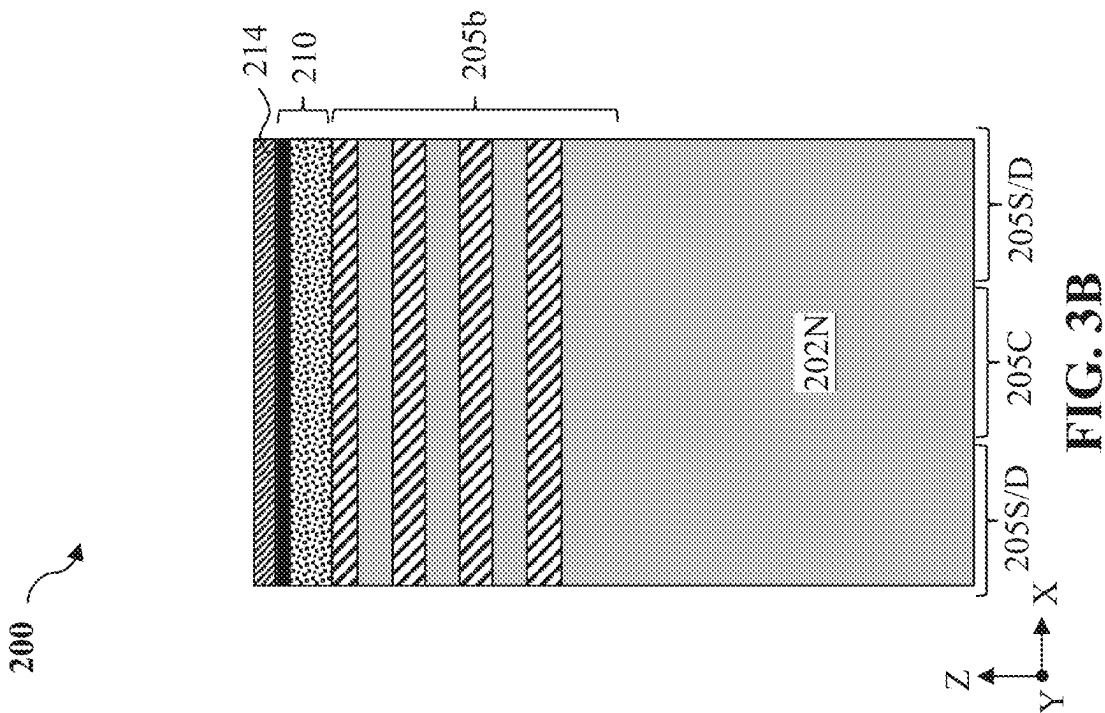


FIG. 1



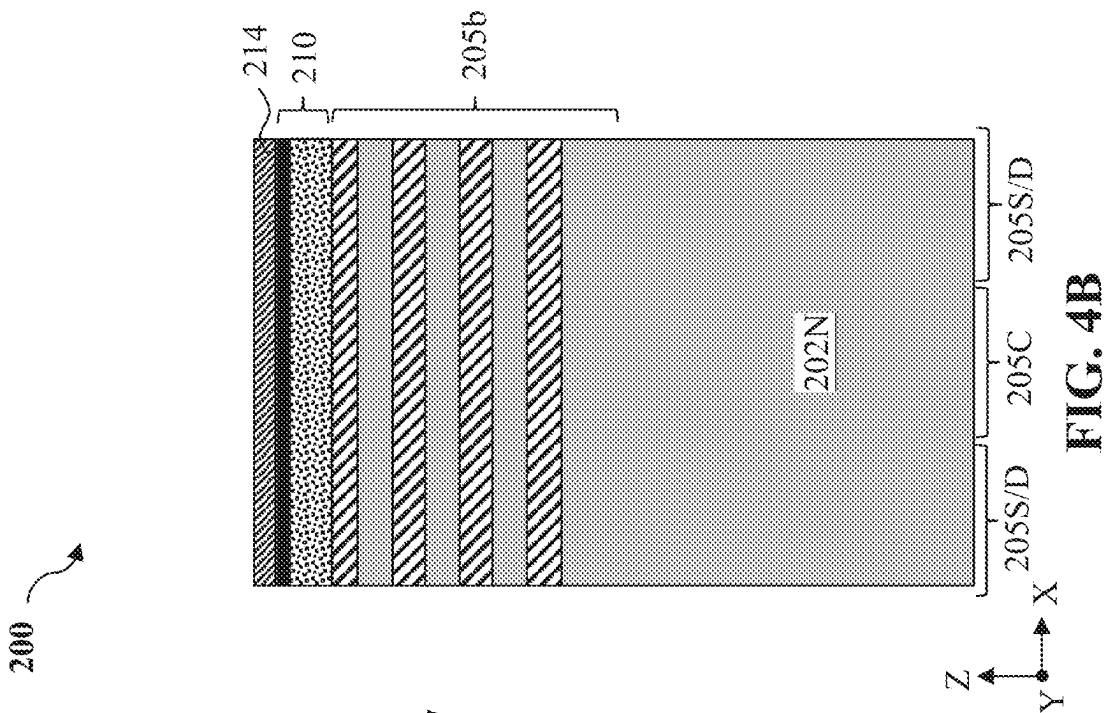


FIG. 4B

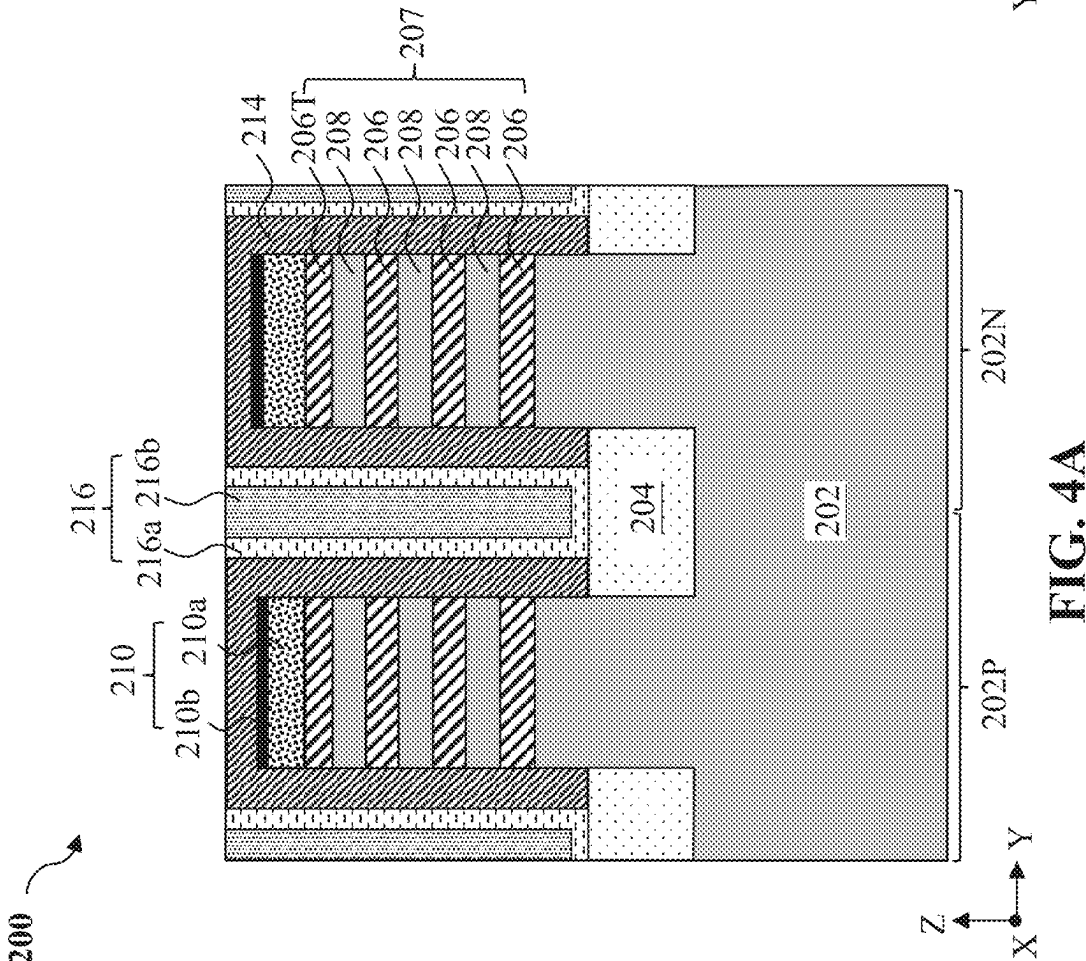
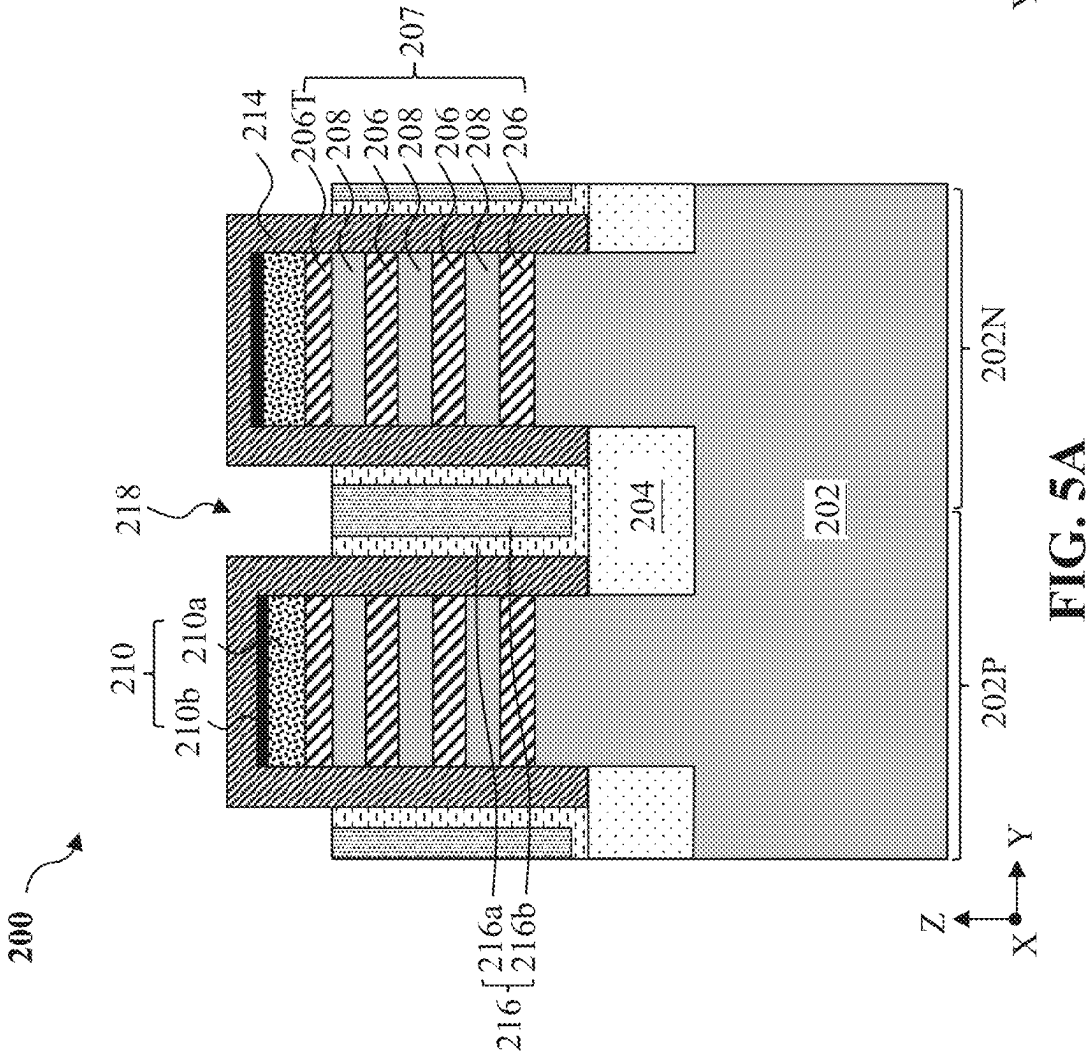
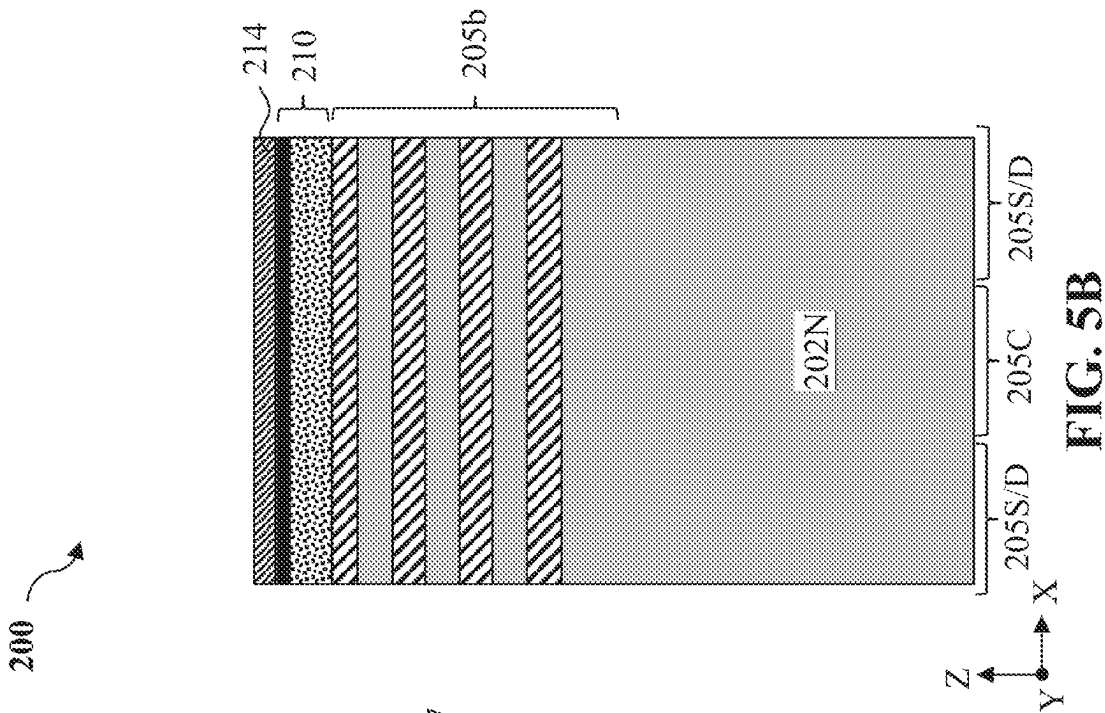


FIG. 4A



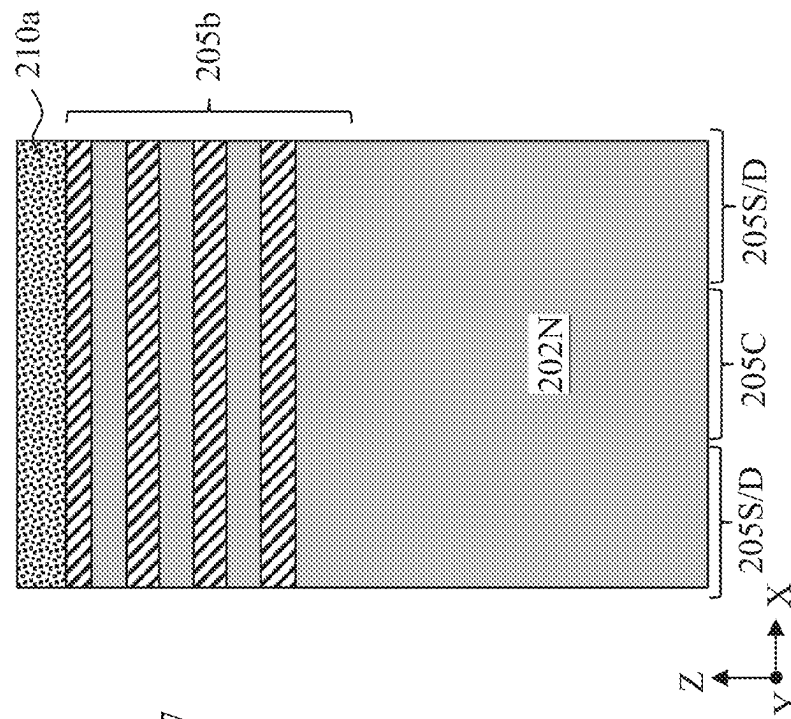
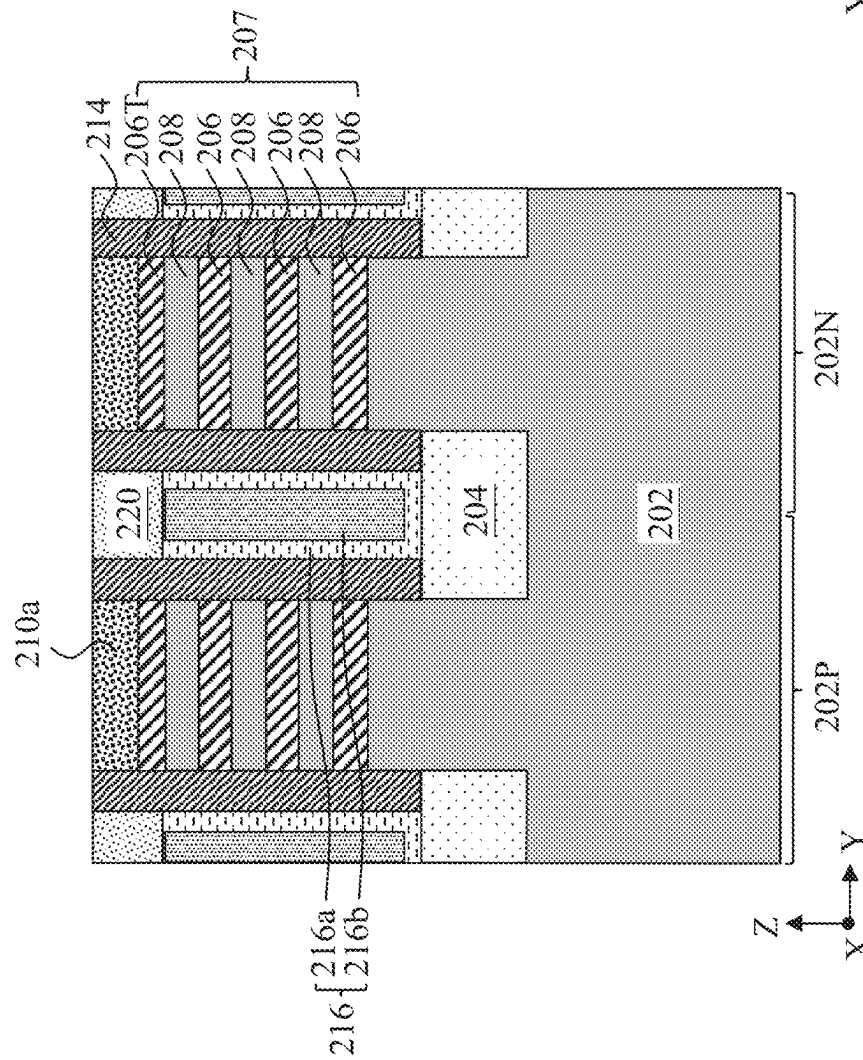
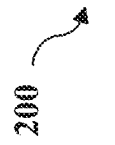
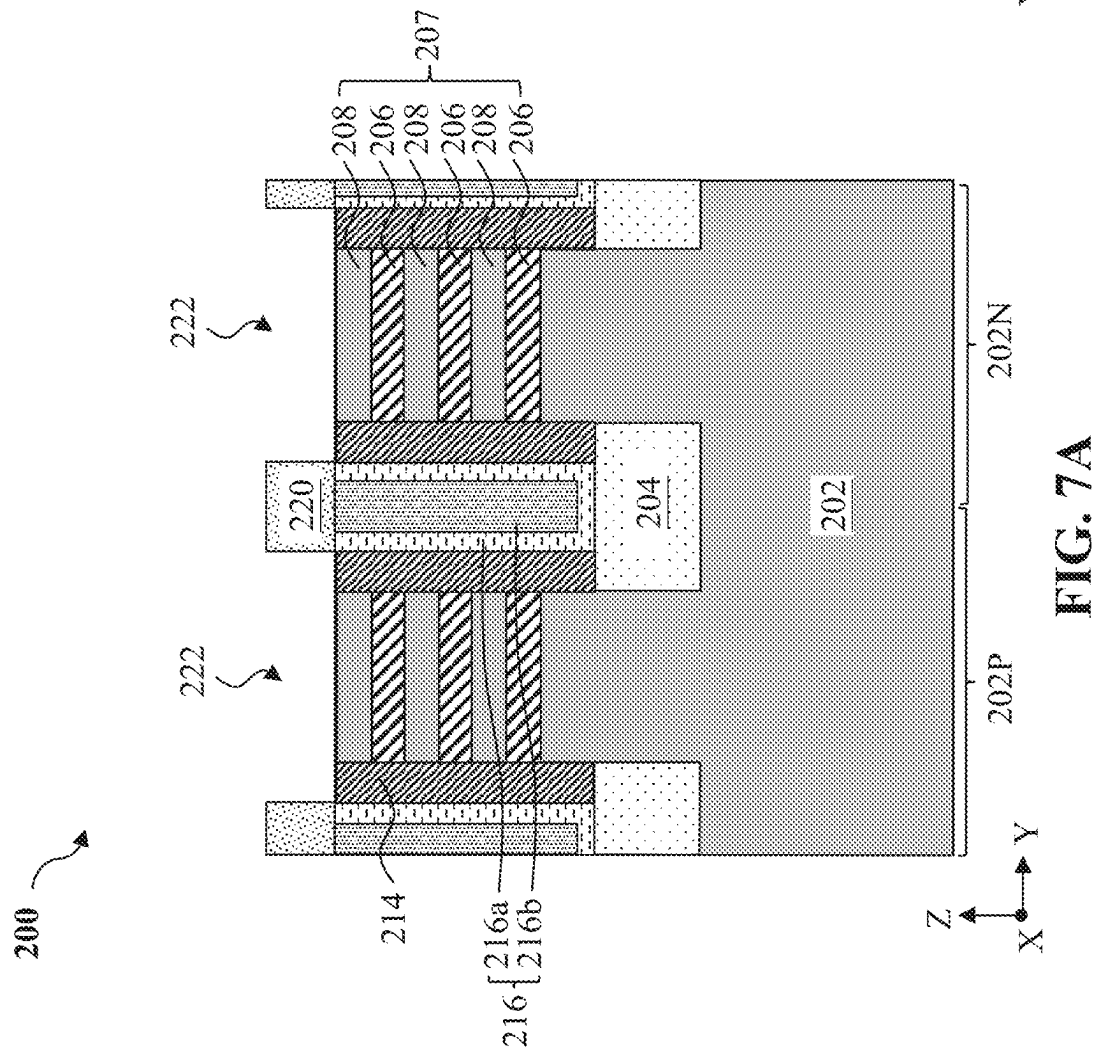
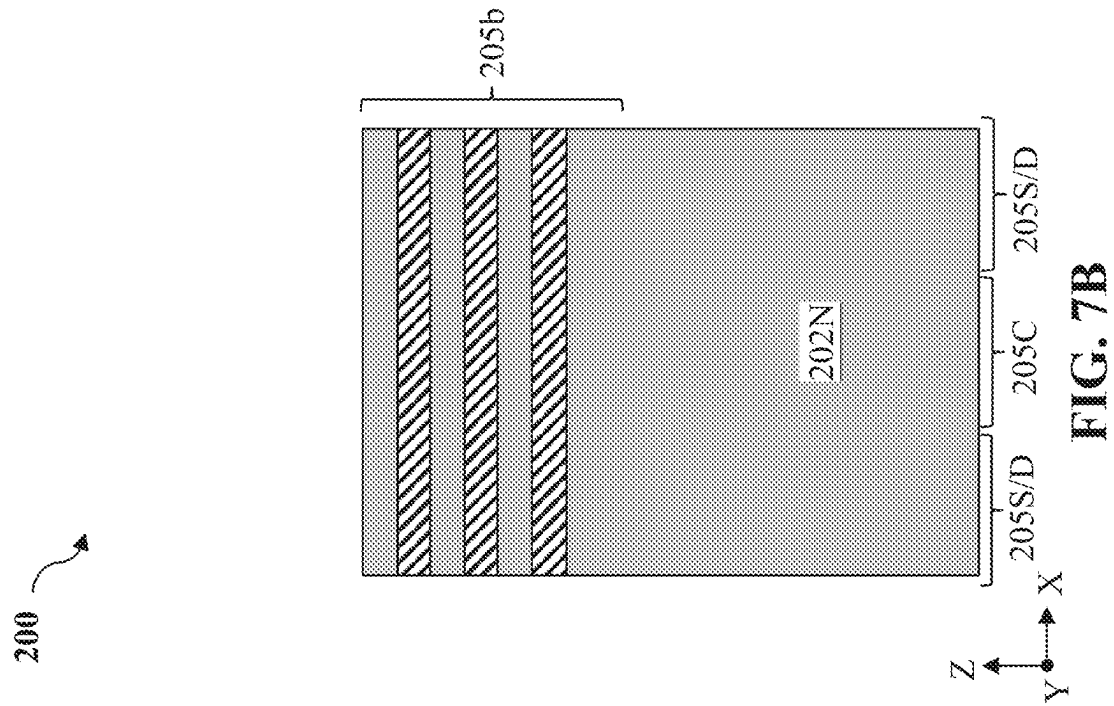


FIG. 3

FIG. 6A



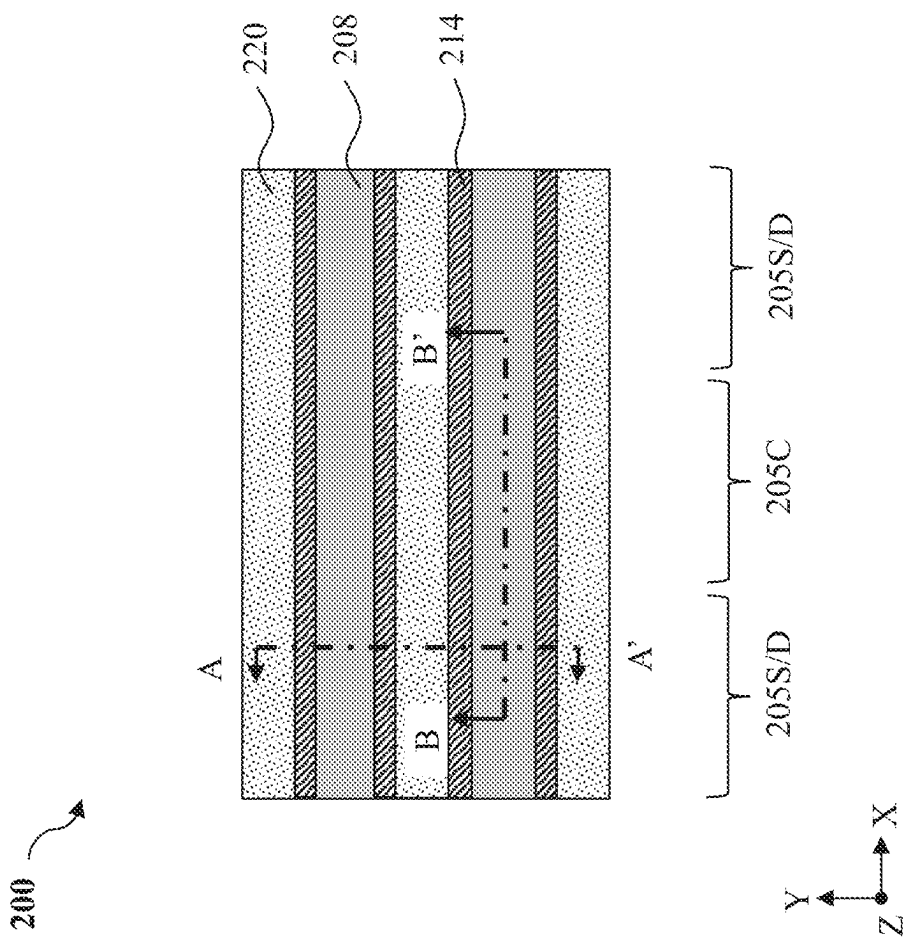


FIG. 7C

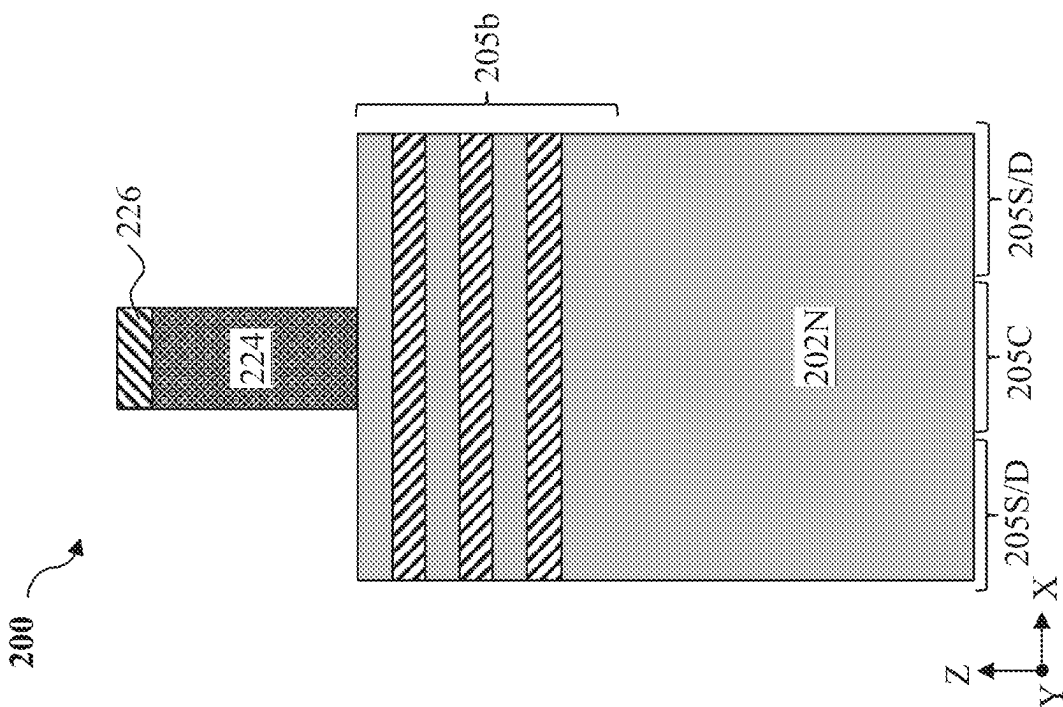


FIG. 8A

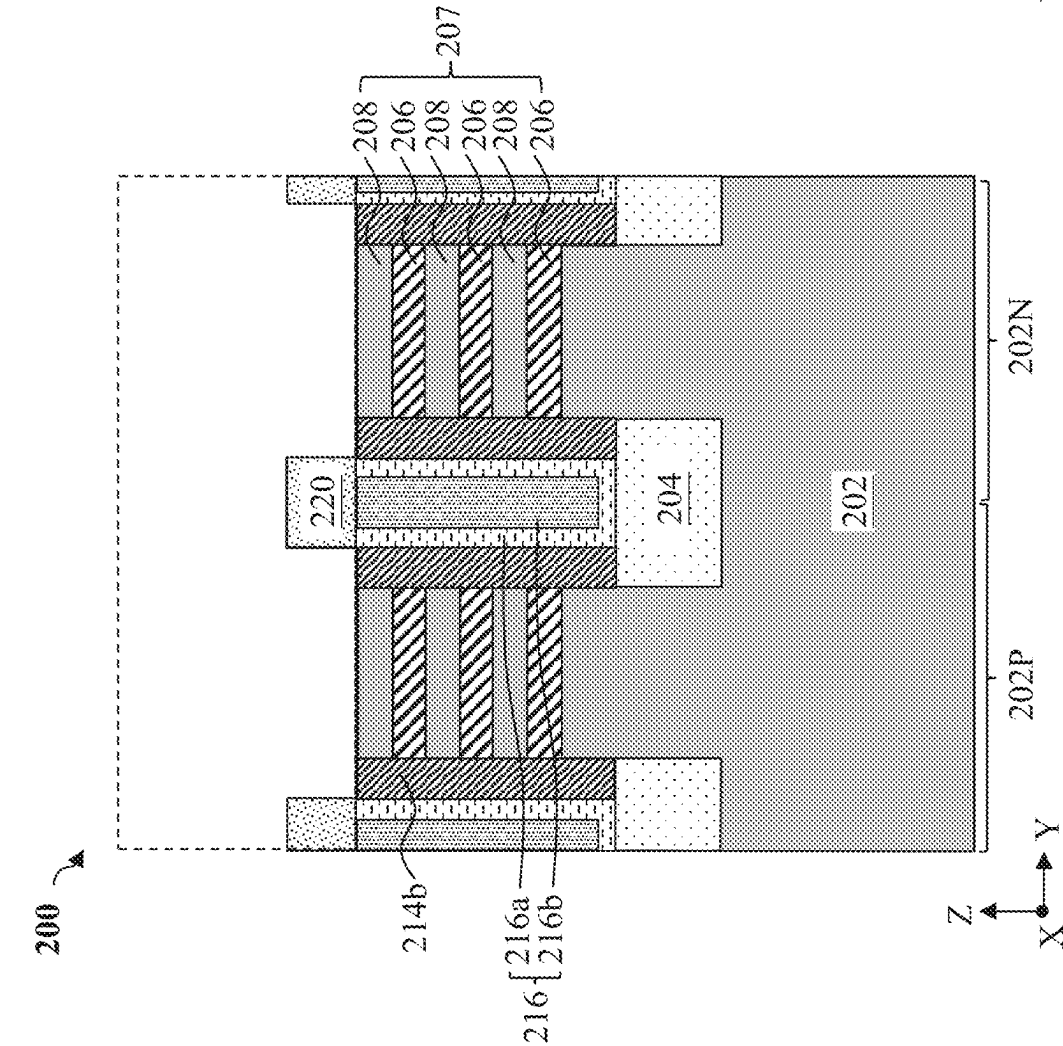


FIG. 8B

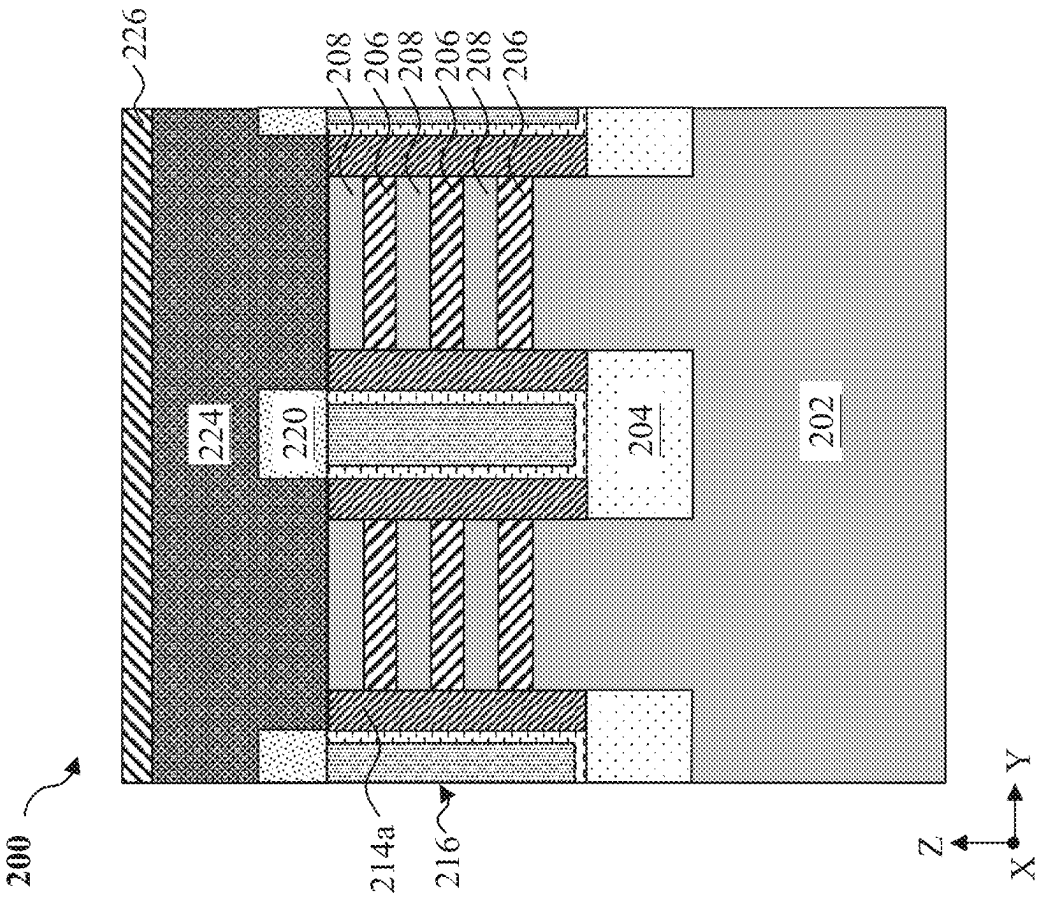


FIG. 8D

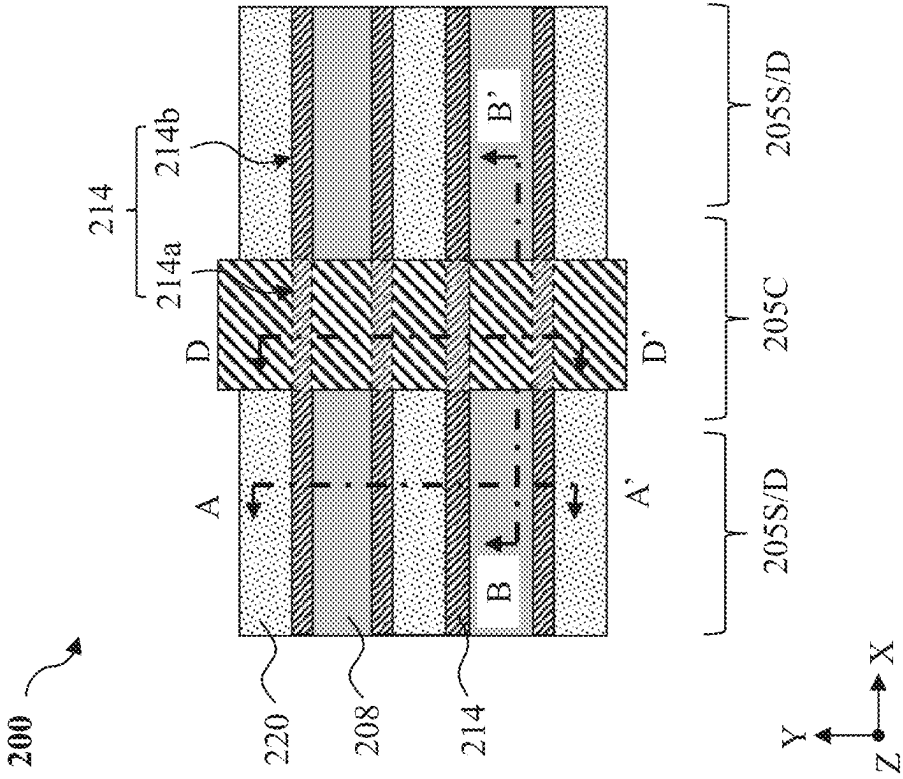
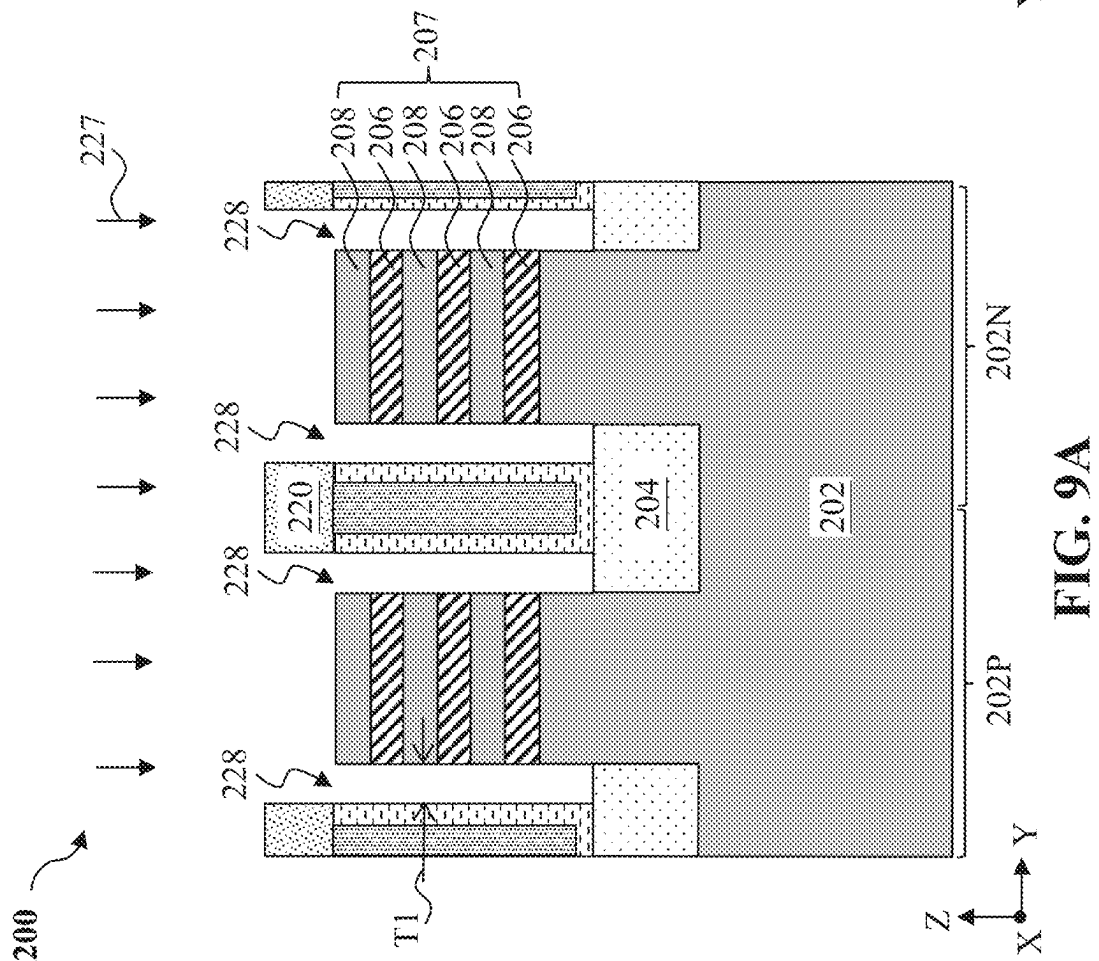
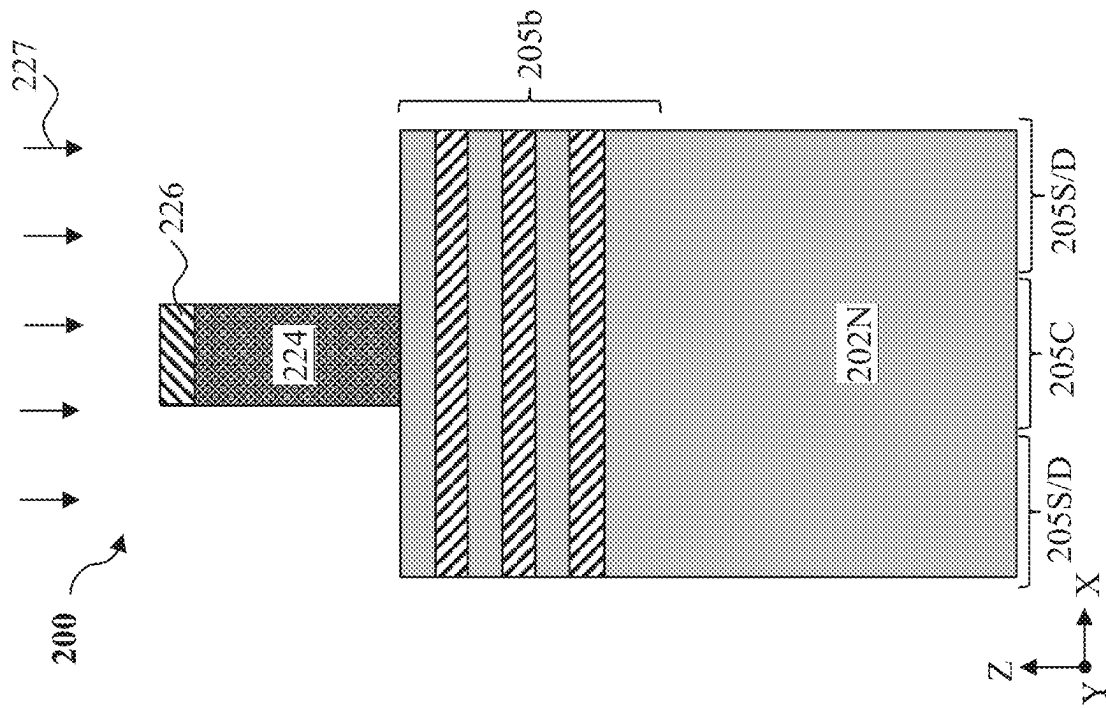


FIG. 8C



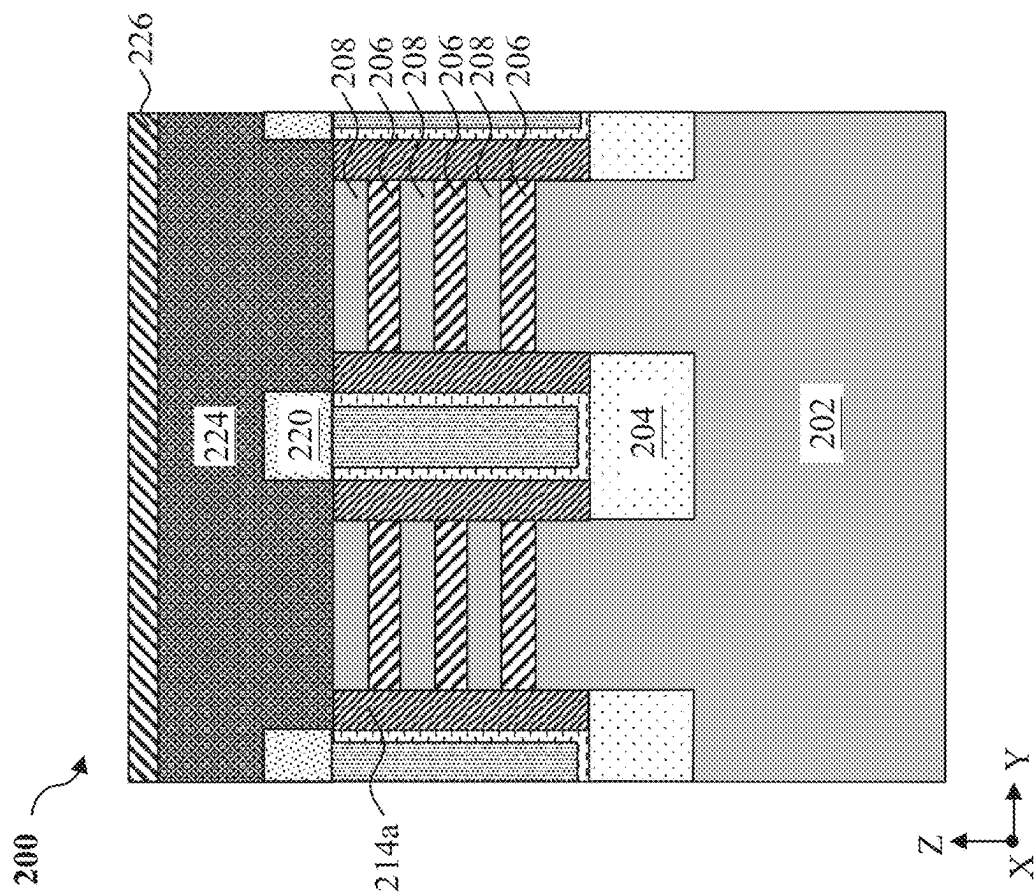


FIG. 9D

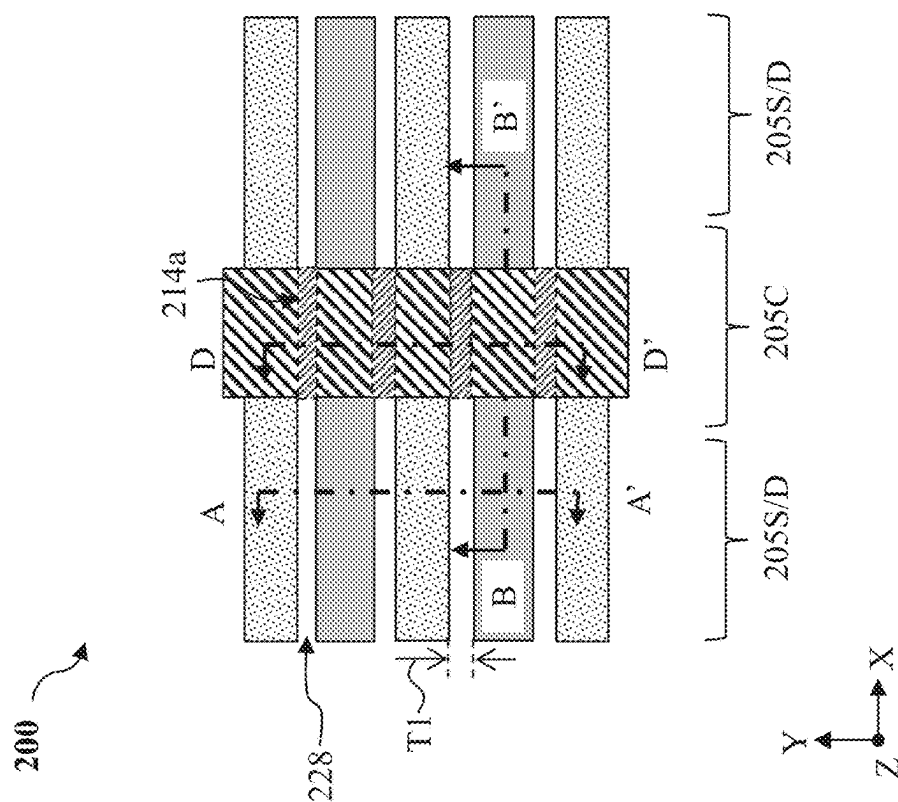
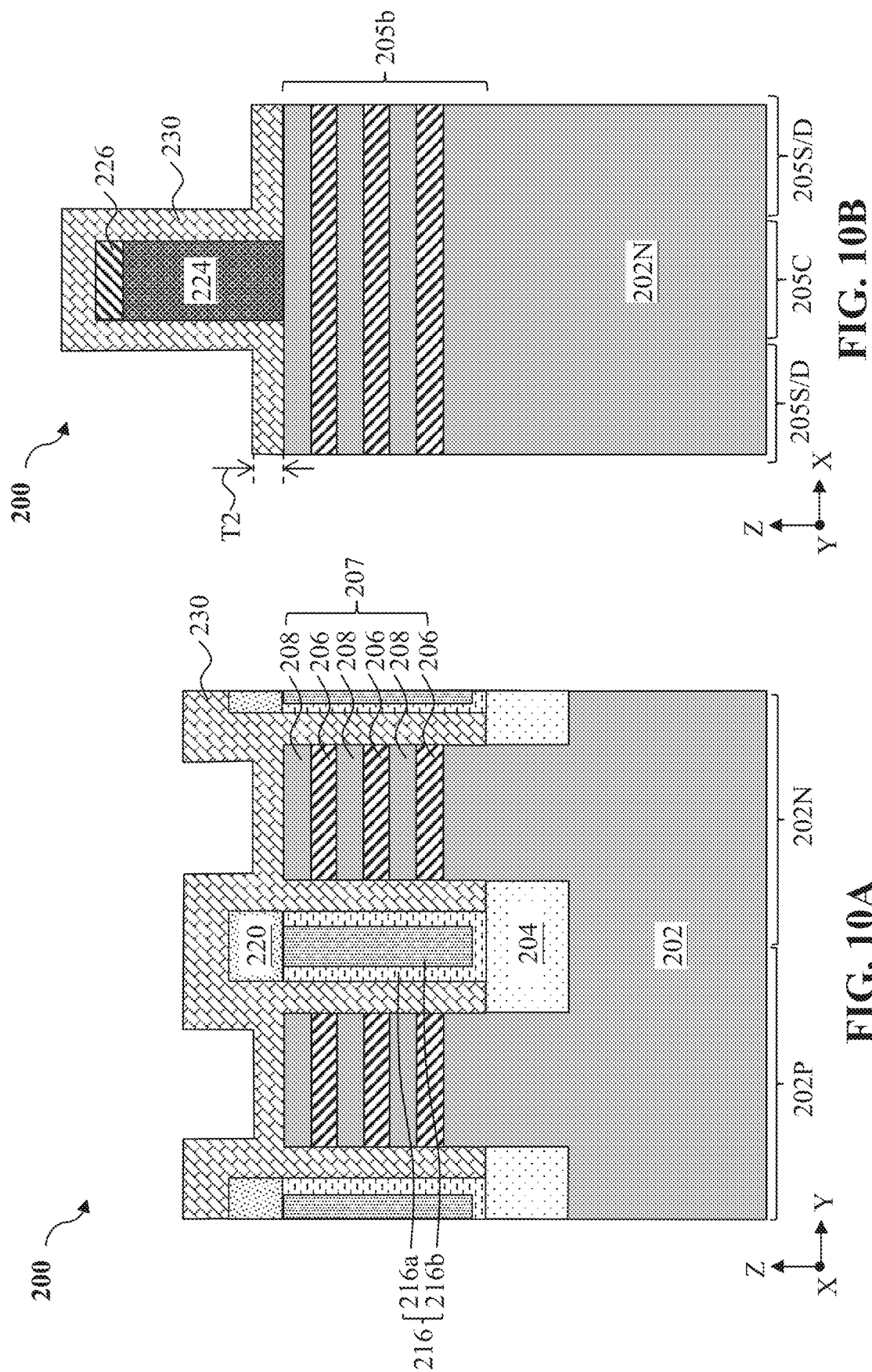
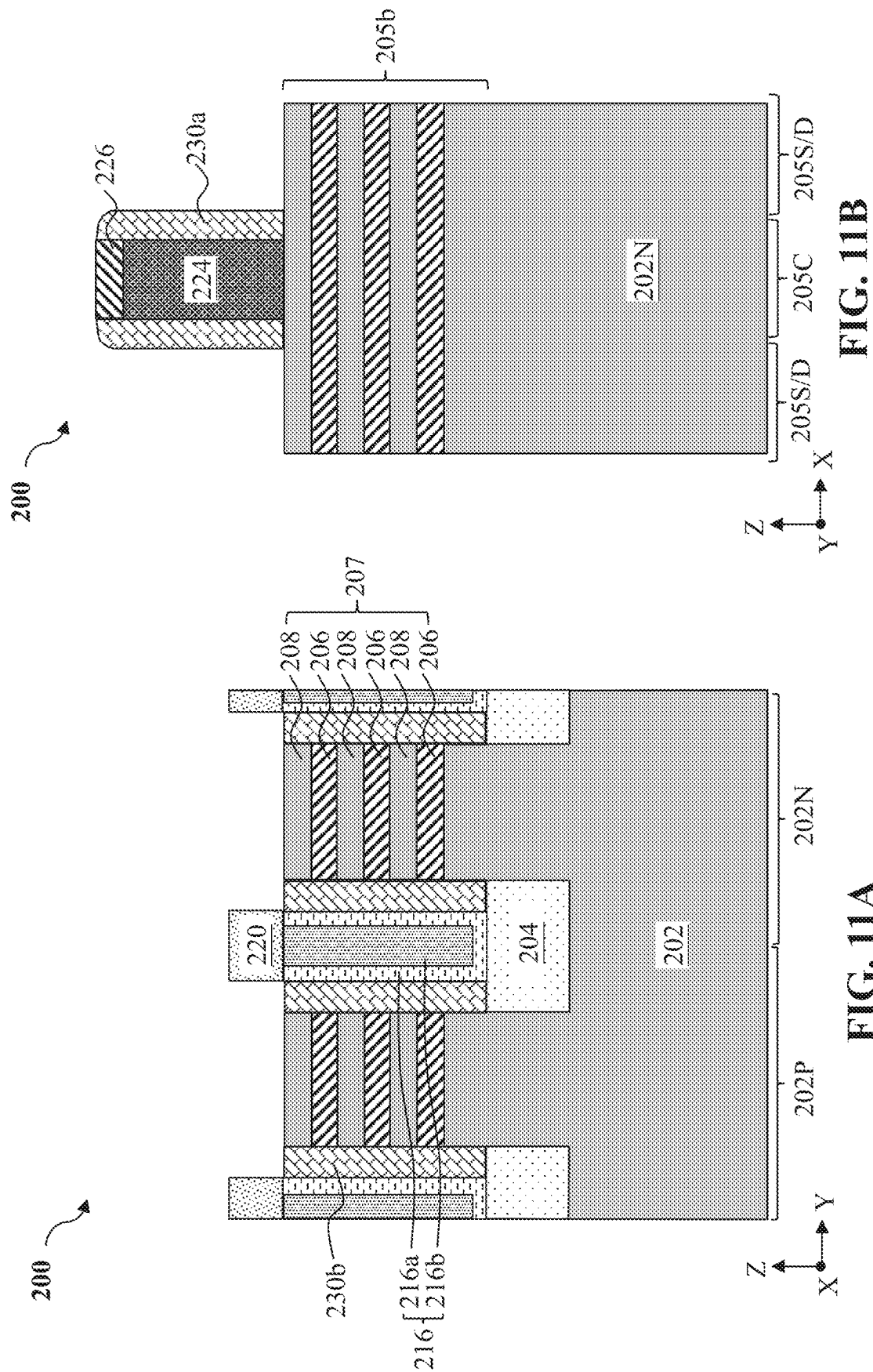


FIG. 9C





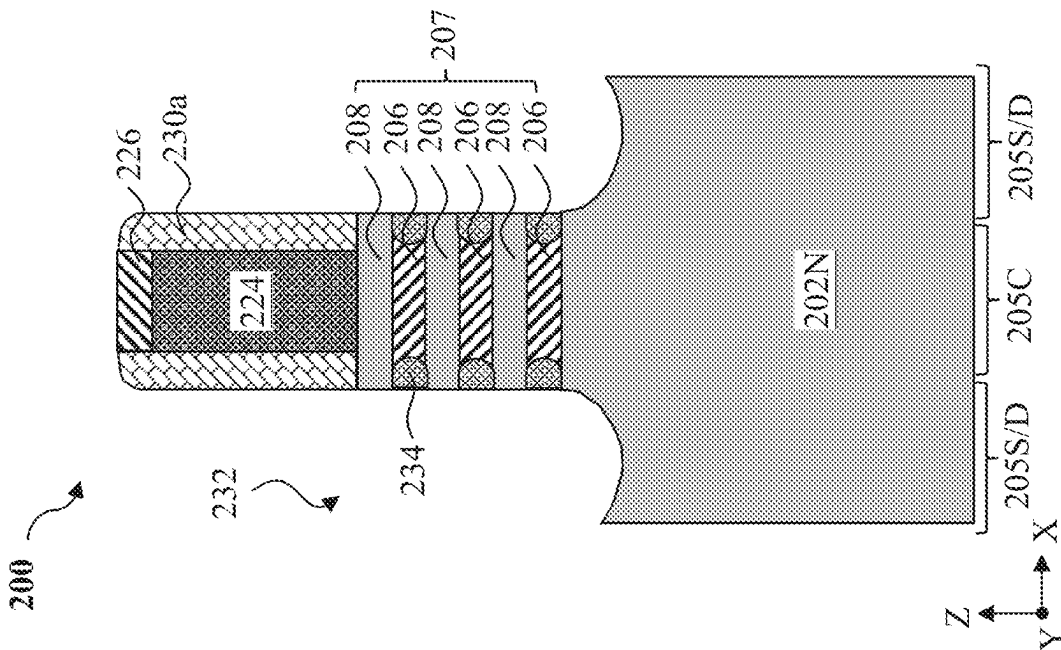


FIG. 12A

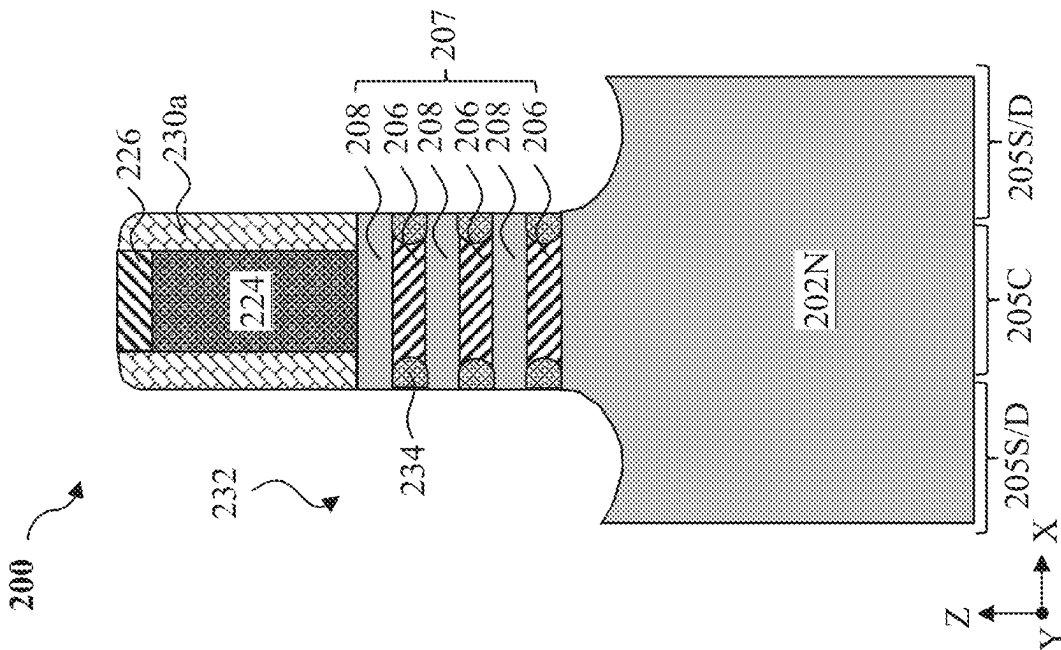


FIG. 12B

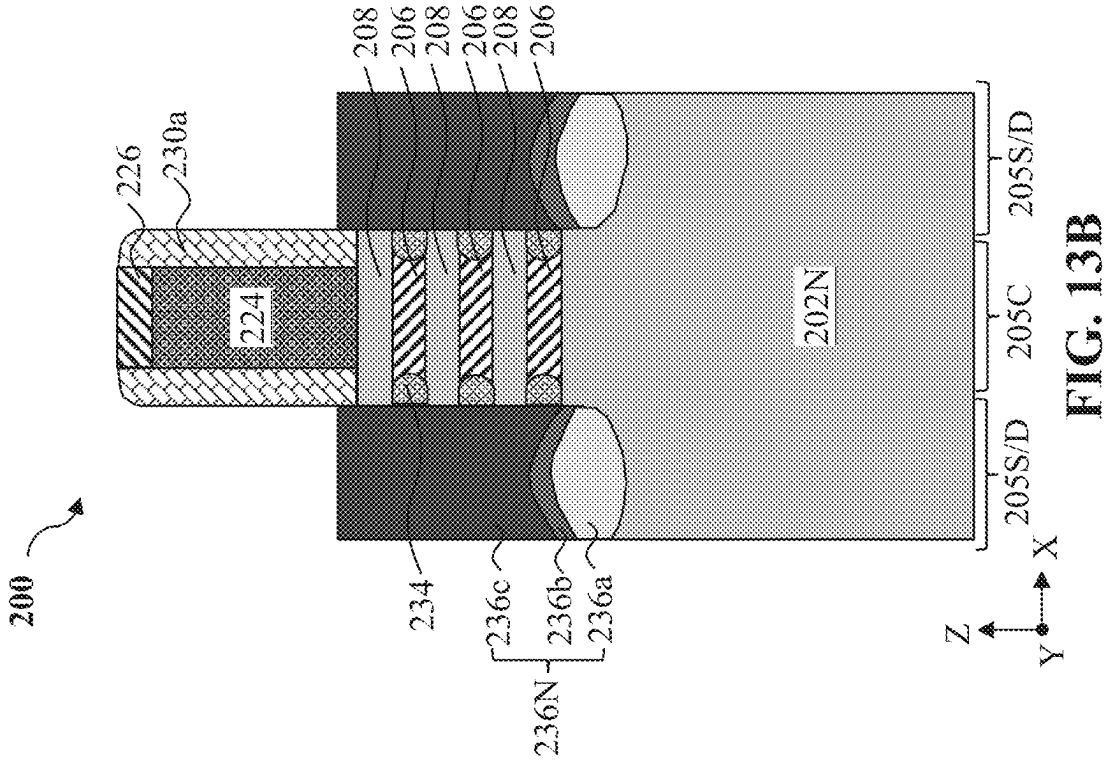


FIG. 13A

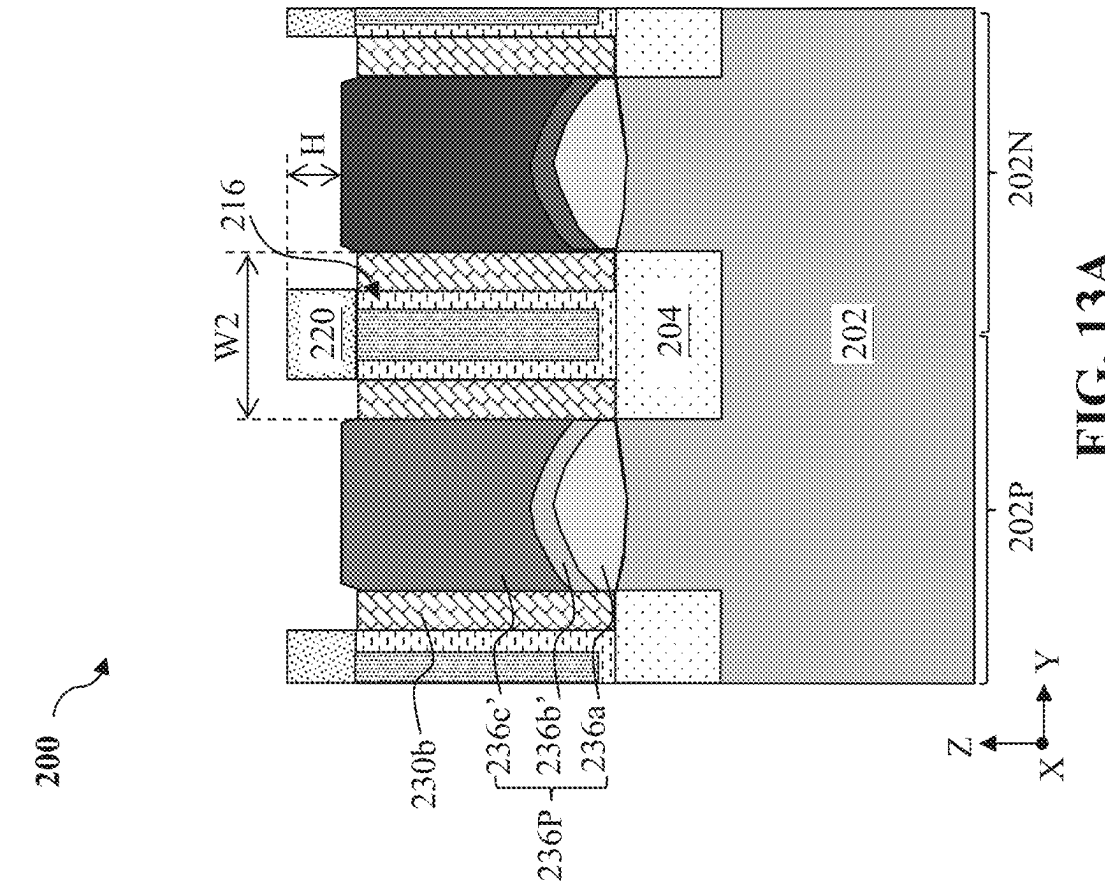


FIG. 13B

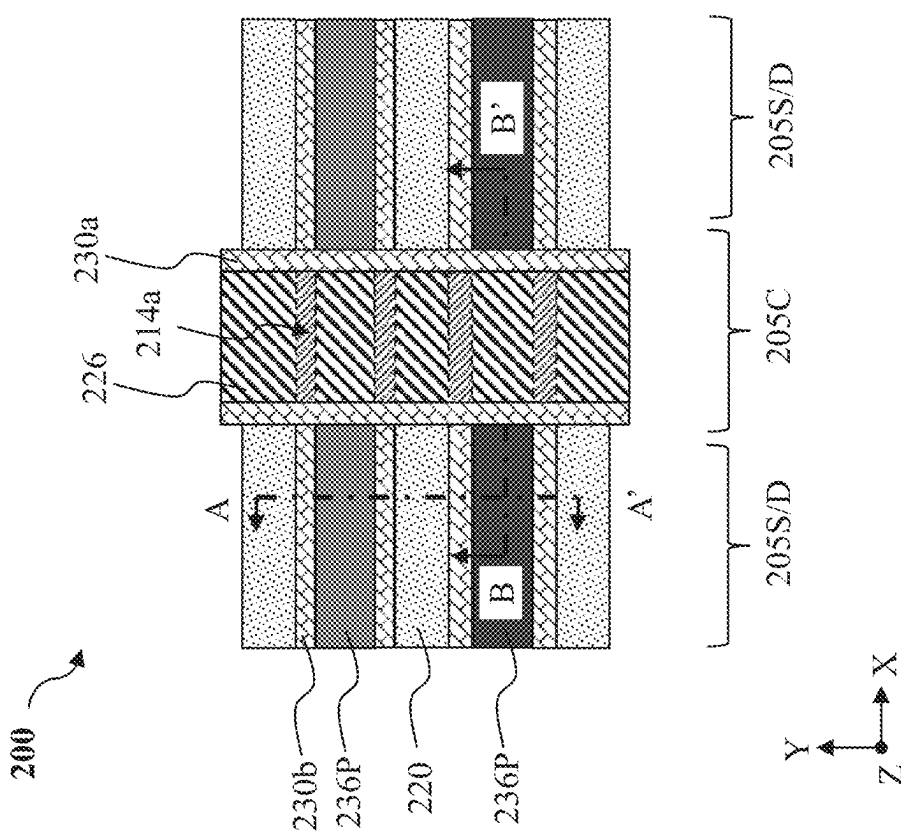


FIG. 13C

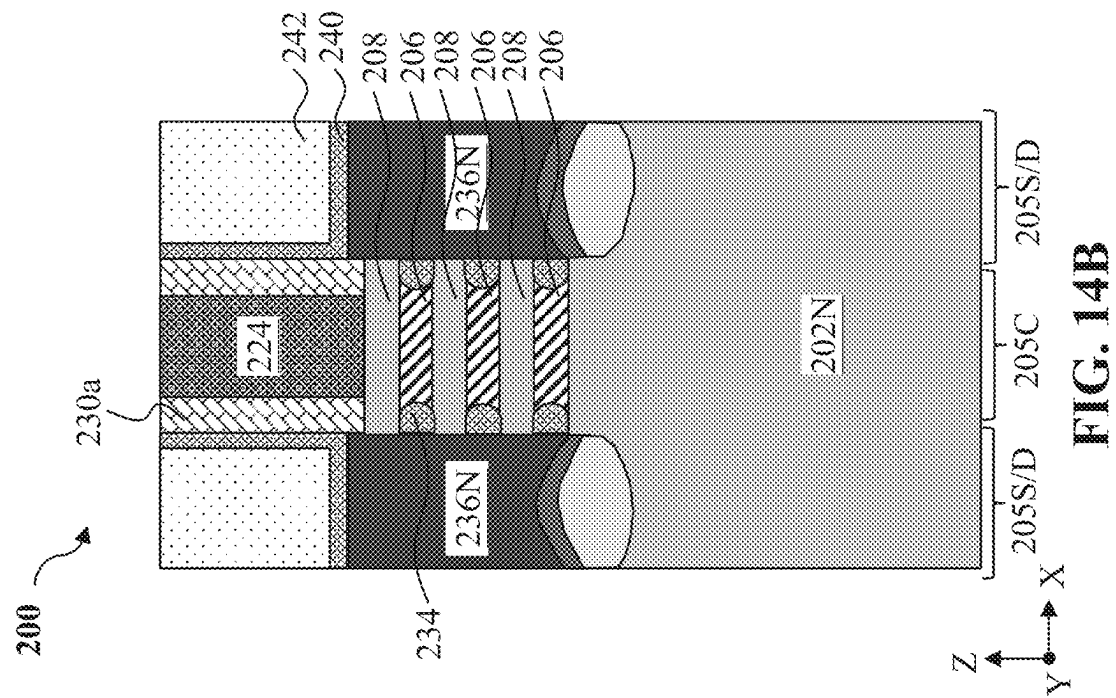


FIG. 14B

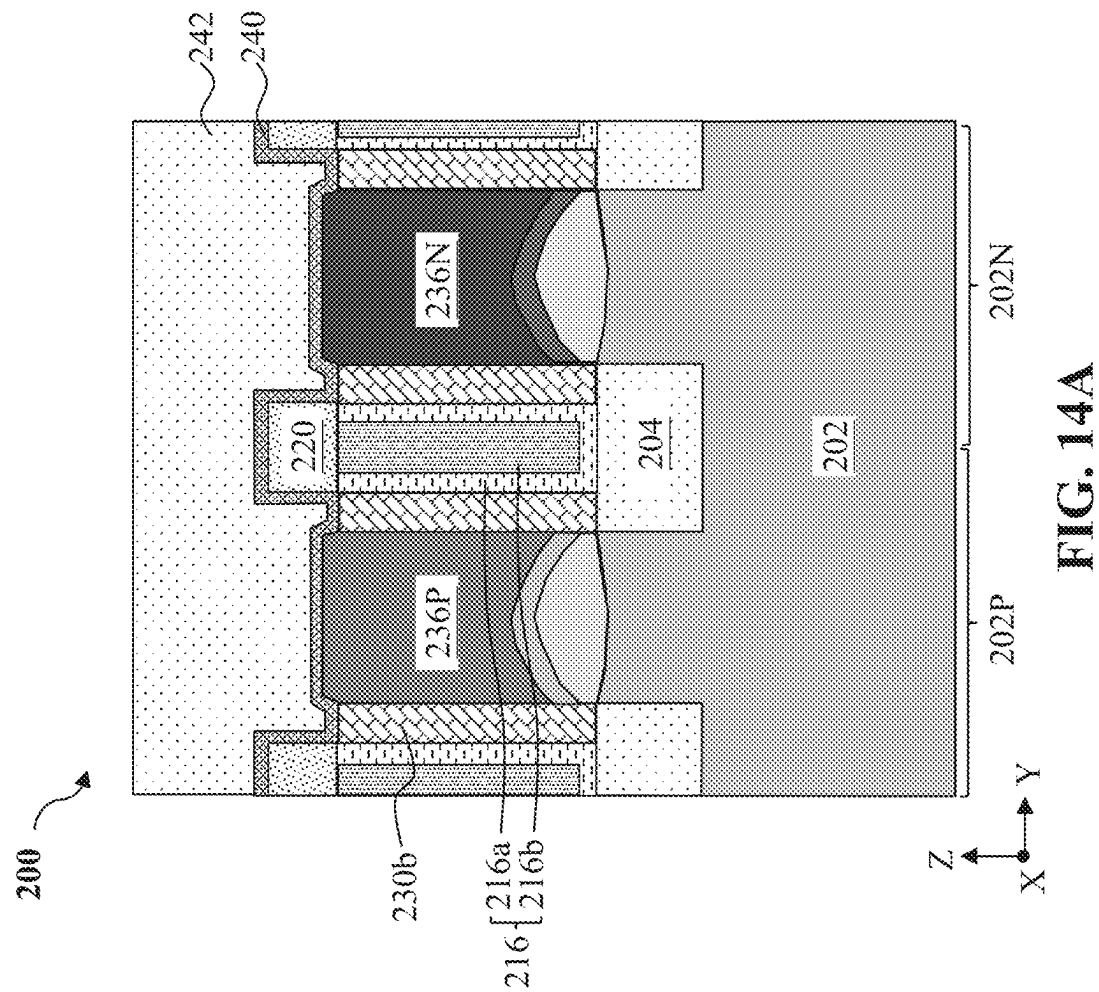
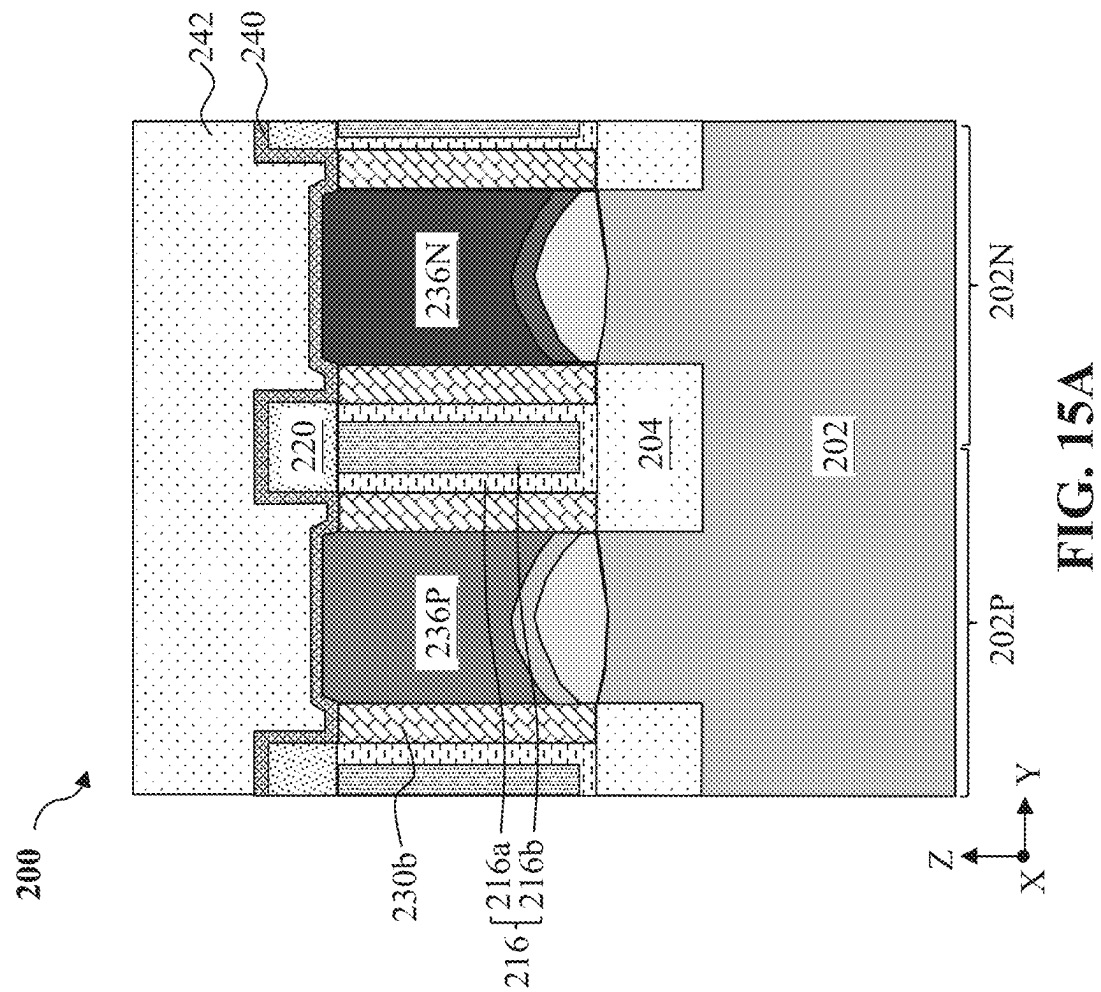
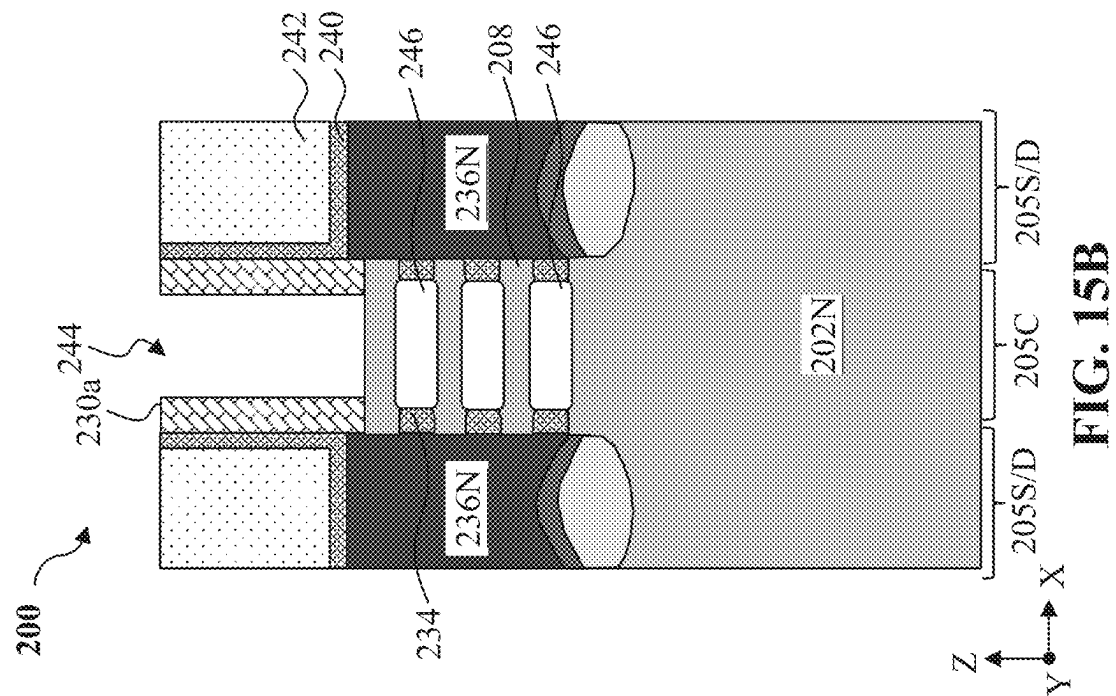


FIG. 14A



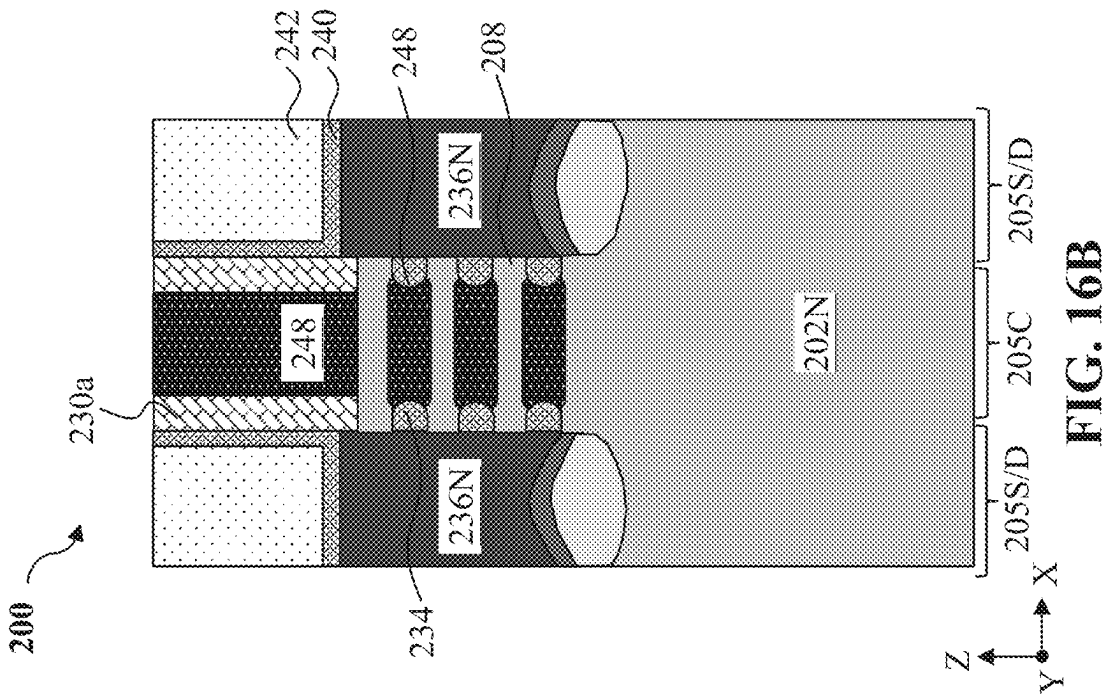


FIG. 16A

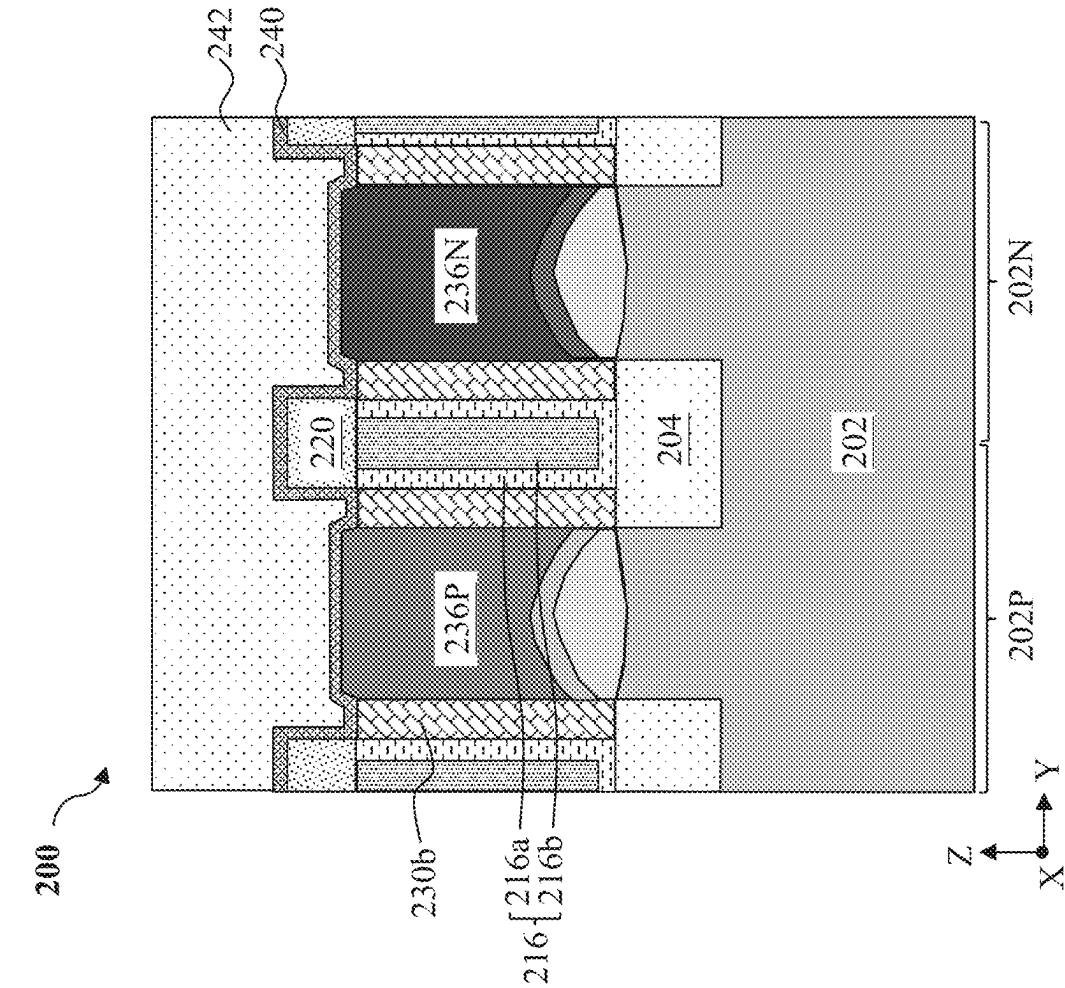


FIG. 16B

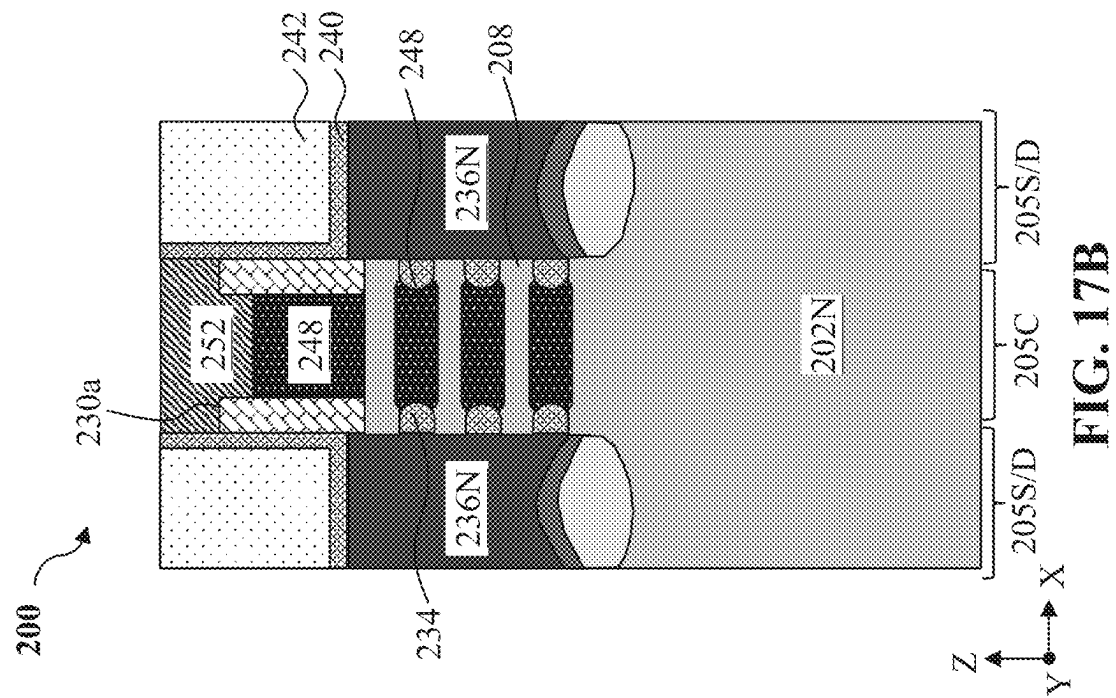


FIG. 17B

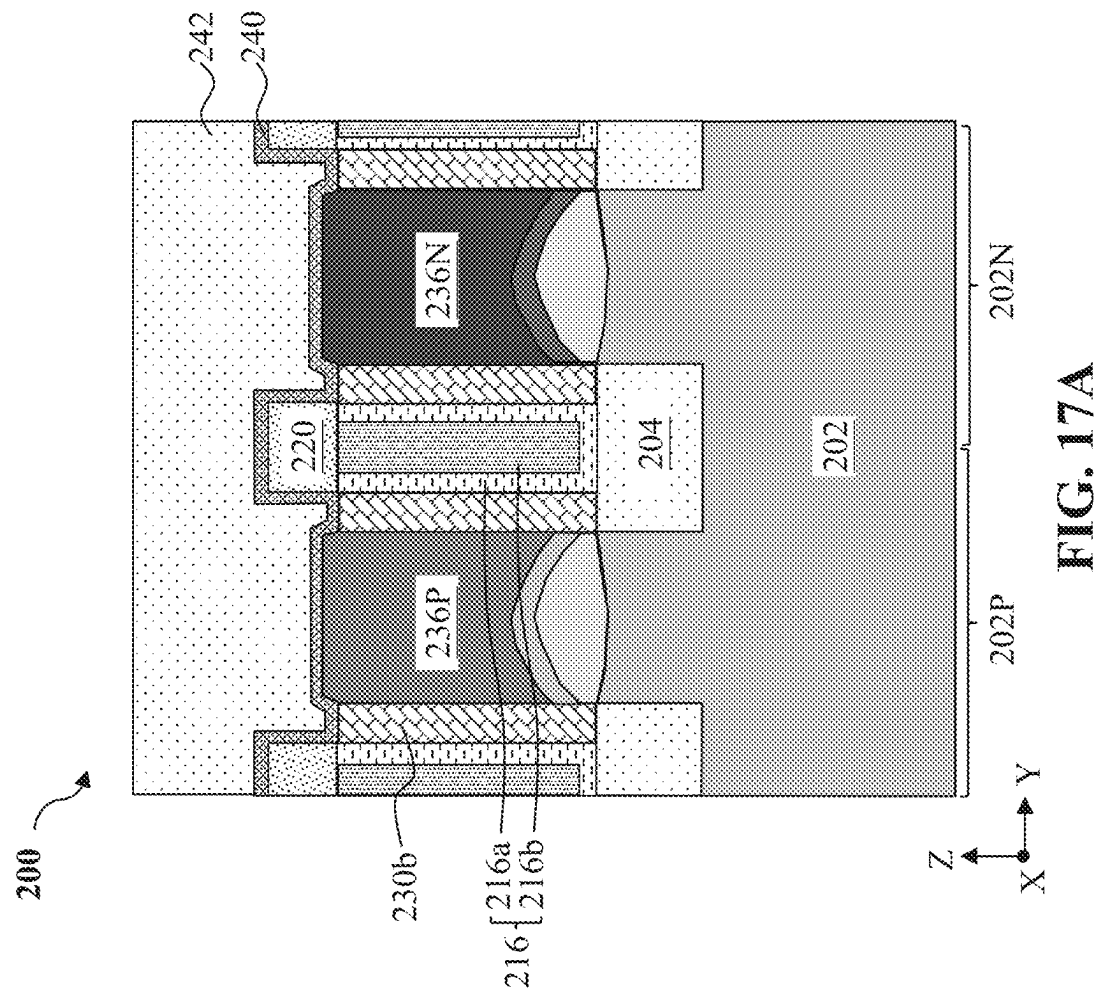


FIG. 17A

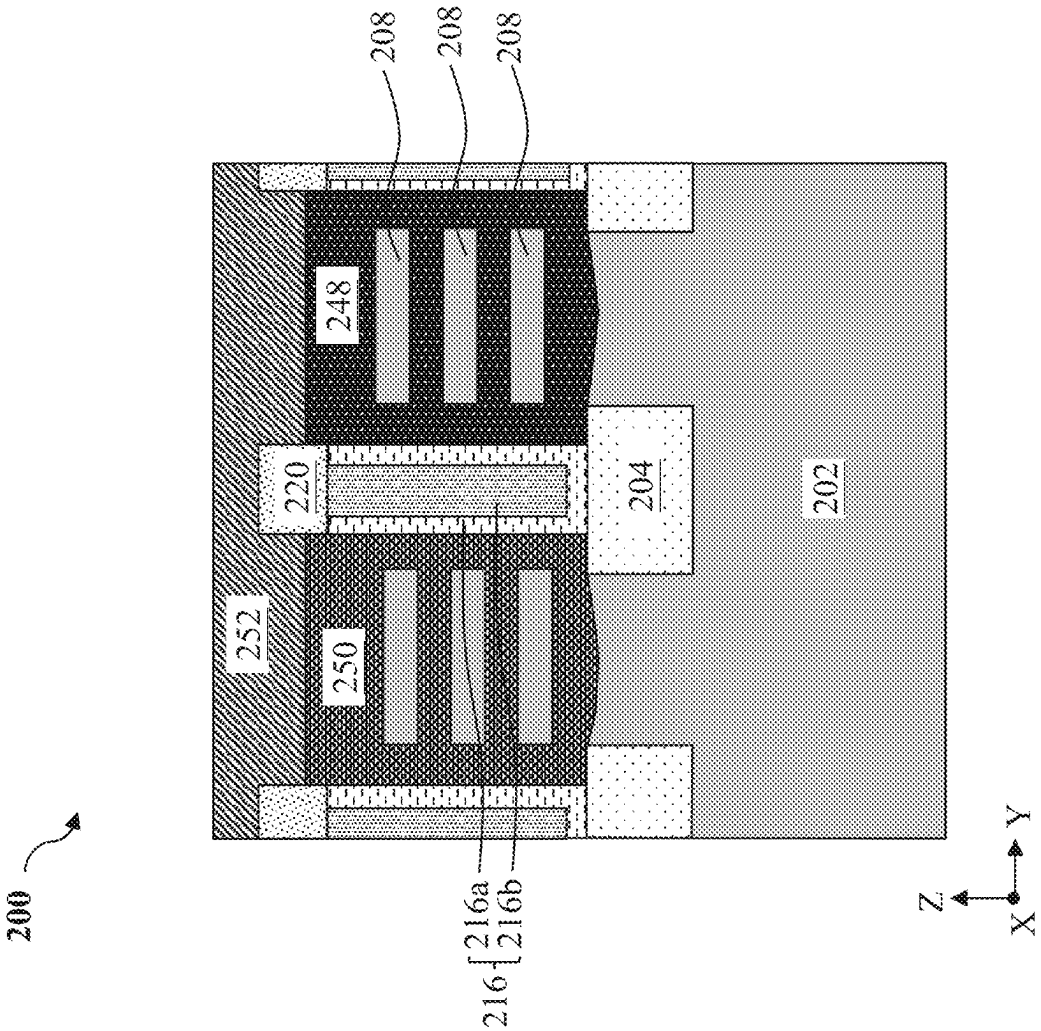


FIG. 17C

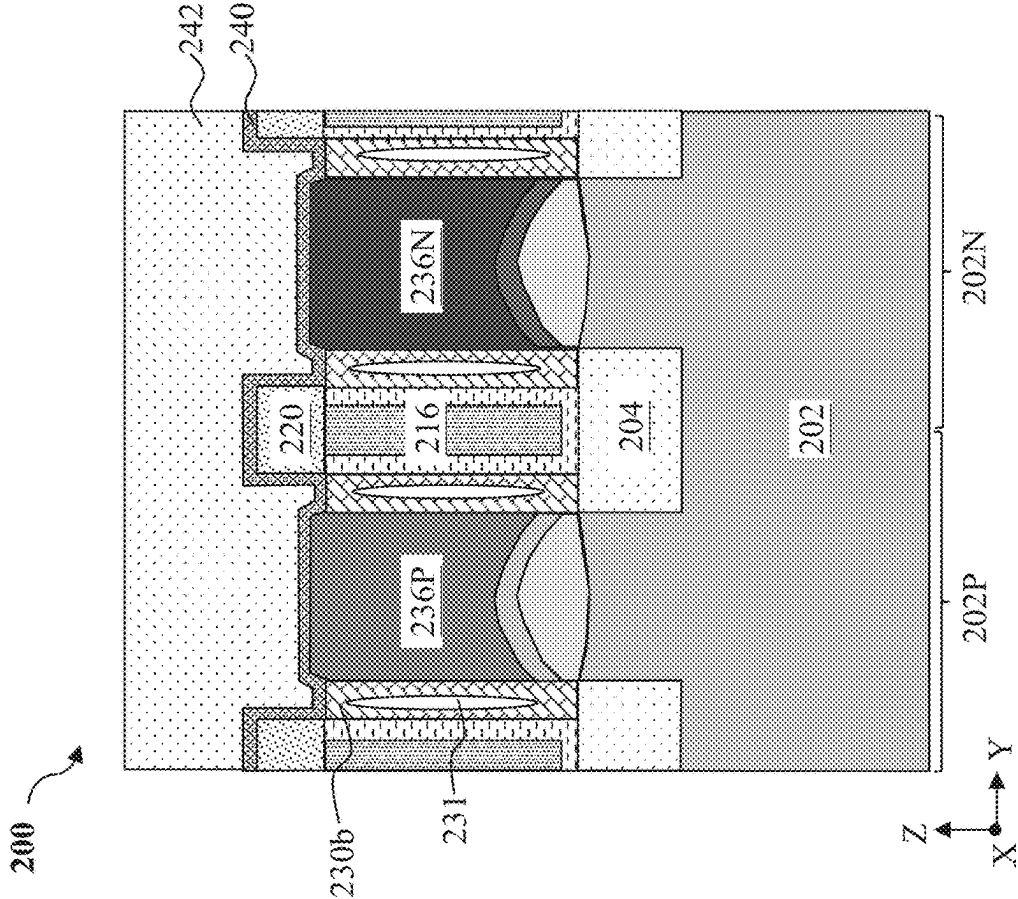


FIG. 18

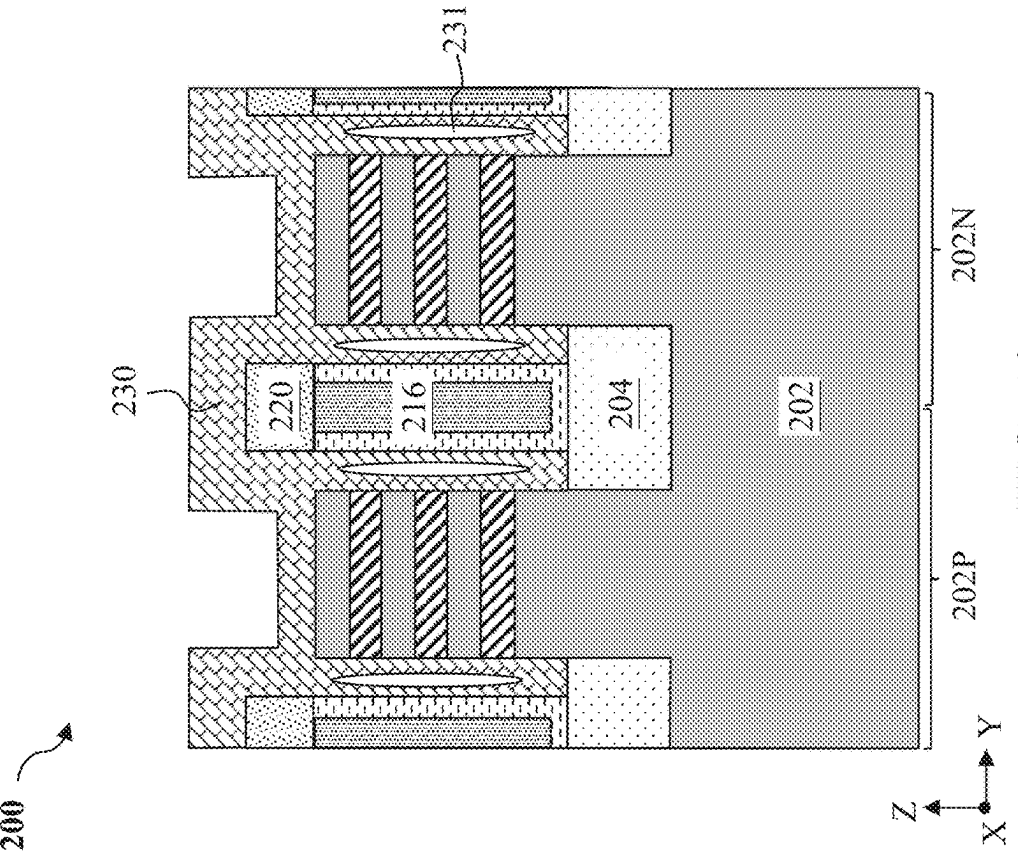


FIG. 19

200

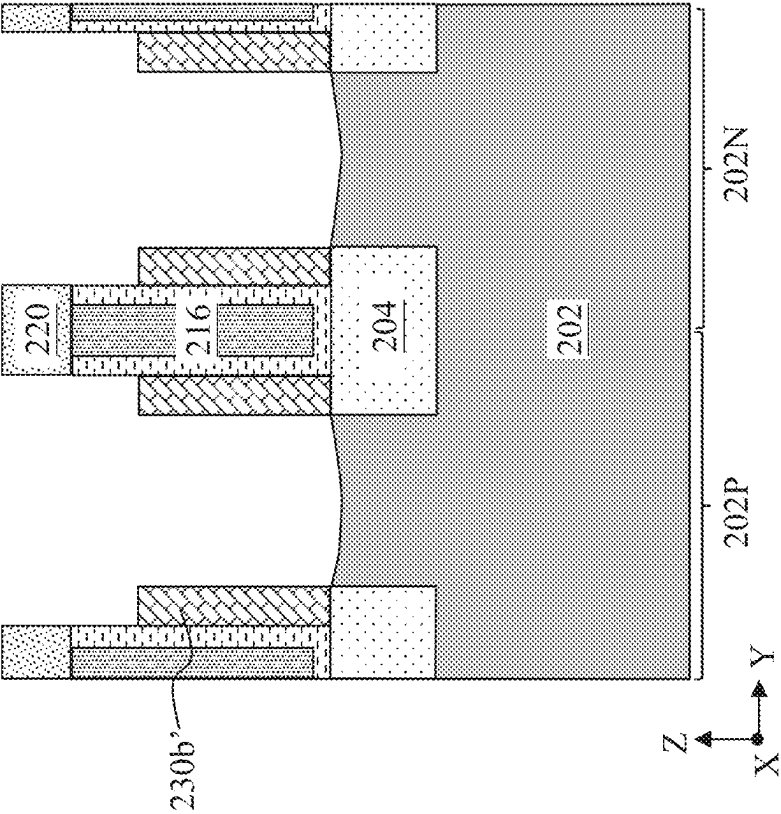


FIG. 20

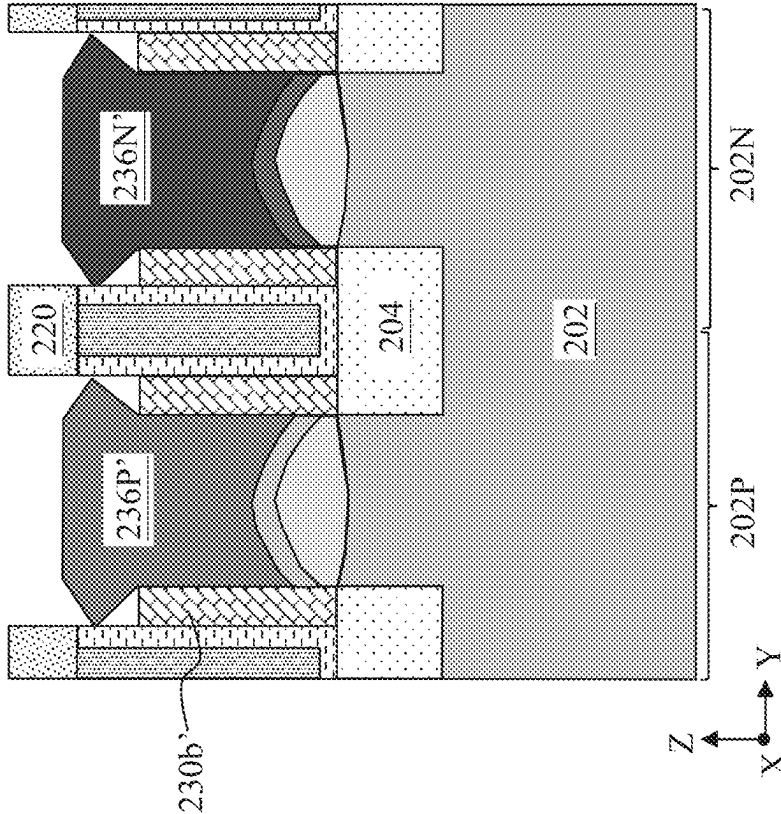


FIG. 21

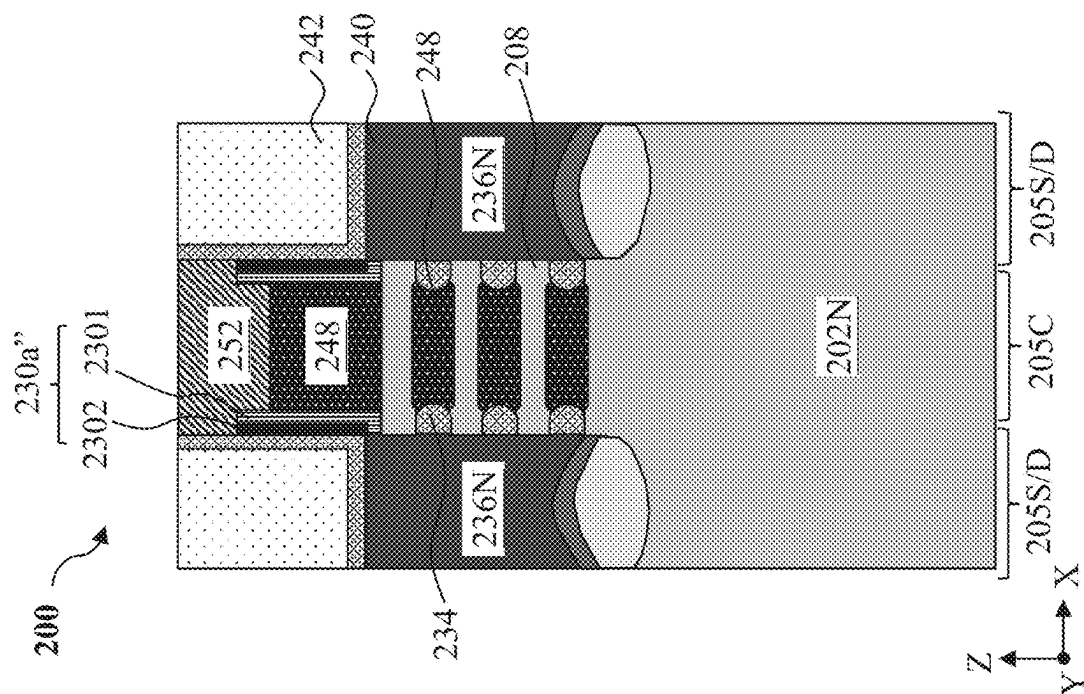


FIG. 22A

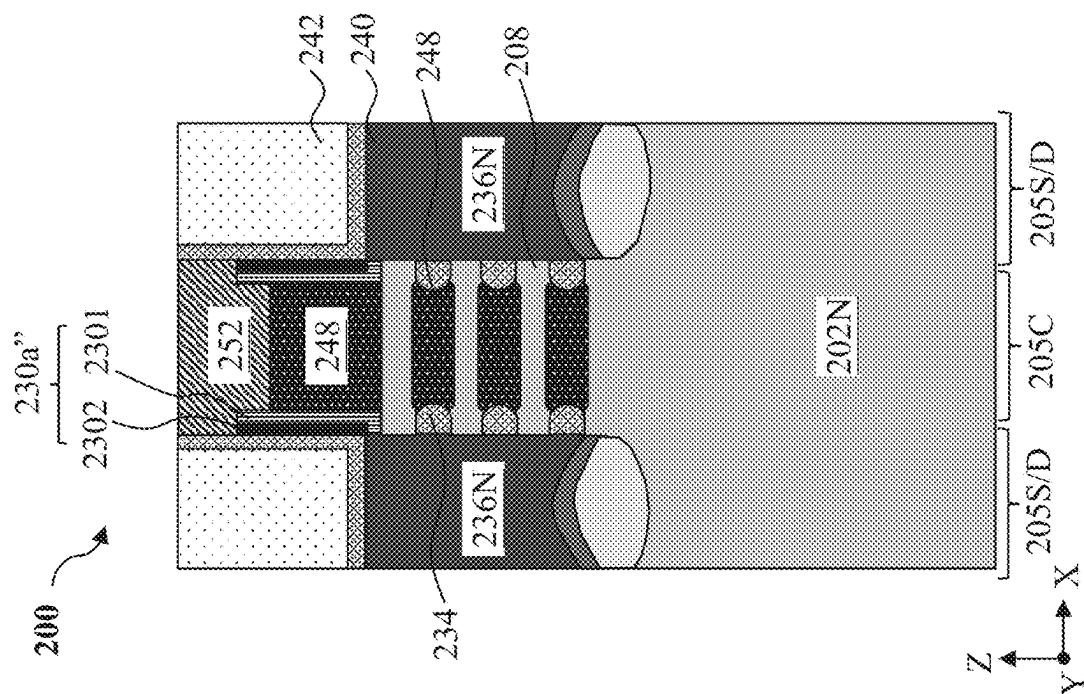


FIG. 22B

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SEMICONDUCTOR DEVICES AND METHODS OF FABRICATING THE SAME

BACKGROUND

The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of processing and manufacturing ICs.

However, as semiconductor devices continue to scale down, challenges arise in achieving desired density and performance. For example, due to densely spaced active regions and the lateral growth of source/drain features, two adjacent source/drain features may merge, leading to unintentional electrical shorting. If electrical shorting occurs, it may degrade device performance and/or lead to device failures. In addition, due to the densely spaced active regions, parasitic capacitance may impact the overall performance of an IC device. In some examples, high parasitic capacitance may lead to lower device speed (e.g., RC delays) when separation distances between the active device regions are reduced to meet design requirements of smaller technology nodes. Accordingly, although existing devices and methods for fabricating such have been generally adequate for their intended purposes, they have not been entirely satisfactory in all respects.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 illustrates a flowchart of an exemplary method for fabricating a semiconductor device, according to various embodiments of the present disclosure.

FIGS. 2A, 3A, 4A, 5A, 6A, 7A, 8A, 9A, 10A, 11A, 12A, 13A, 14A, 15A, 16A, and 17A (FIGS. 2A-17A) illustrate cross-sectional views of a workpiece during various fabrication stages in the method of FIG. 1, according to one or more aspects of the present disclosure.

FIGS. 2B, 3B, 4B, 5B, 6B, 7B, 8B, 9B, 10B, 11B, 12B, 13B, 14B, 15B, 16B, and 17B (FIGS. 2B-17B) illustrate fragmentary cross-sectional views of the workpiece taken along line B-B' shown in FIG. 2A during various fabrication stages in the method of FIG. 1, according to one or more aspects of the present disclosure.

FIGS. 7C, 8C, 9C (FIGS. 7C-9C) and 13C illustrate fragmentary planar top views of a workpiece during various fabrication stages in the method of FIG. 1, according to various aspects of the disclosure.

FIGS. 8D, 9D (FIGS. 8D-9D) and 17C illustrate fragmentary cross-sectional views of the workpiece taken along

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line D-D' shown in FIG. 8C during various fabrication stages in the method of FIG. 1, according to one or more aspects of the present disclosure.

FIGS. 18 and 19 illustrate cross-sectional views of a workpiece during a fabrication process according to the method of FIG. 1, according to a first alternative embodiment and one or more aspects of the present disclosure.

FIGS. 20 and 21 (FIGS. 18-21) illustrate cross-sectional views of a workpiece during a fabrication process according to the method of FIG. 1, according to a second alternative embodiment and one or more aspects of the present disclosure.

FIGS. 22A and 22B illustrate fragmentary cross-sectional views of a workpiece during various fabrication stages in the method of FIG. 1, according to a third alternative embodiment and one or more aspects of the present disclosure.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

Further, when a number or a range of numbers is described with “about,” “approximate,” and the like, the term is intended to encompass numbers that are within a reasonable range considering variations that inherently arise during manufacturing as understood by one of ordinary skill in the art. For example, the number or range of numbers encompasses a reasonable range including the number described, such as within $\pm 10\%$ of the number described, based on known manufacturing tolerances associated with manufacturing a feature having a characteristic associated with the number. For example, a material layer having a thickness of “about 5 nm” can encompass a dimension range from 4.25 nm to 5.75 nm where manufacturing tolerances associated with depositing the material layer are known to be $\pm 15\%$ by one of ordinary skill in the art. Still further, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

As integrated circuit (IC) technologies progress towards smaller technology nodes, multi-gate devices have been introduced to improve gate control by increasing gate-channel coupling, reducing off-state current, and reducing short-channel effects (SCEs). A multi-gate device generally refers to a device having a gate structure, or portion thereof, disposed over more than one side of a channel region. Fin-like field effect transistors (FinFETs) and multi-bridge-channel (MBC) transistors are examples of multi-gate devices that have become popular and promising candidates for high performance and low leakage applications. A FinFET has an elevated channel wrapped by a gate on more than one side (for example, the gate wraps a top and sidewalls of a “fin” of semiconductor material extending from a substrate). An MBC transistor has a gate structure that can extend, partially or fully, around a channel region to provide access to the channel region on two or more sides. Because its gate structure surrounds the channel regions, an MBC transistor may also be referred to as a surrounding gate transistor (SGT) or a gate-all-around (GAA) transistor. The channel region of an MBC transistor may be formed from nanowires, nanosheets, or other nanostructures and for that reason, an MBC transistor may also be referred to as a nanowire transistor or a nanosheet transistor.

Formation of a FinFET or an MBC transistor includes patterning semiconductor material(s) into fin-shaped active regions, forming a dummy fin between two adjacent fin-shaped active regions to isolate source/drain features to be formed in source/drain trenches. In some embodiments, a helmet layer may be formed on the dummy fin and work together with the dummy fin to electrically divide a functional gate stack into two pieces. As described above, aggressive scaling down of IC dimensions has resulted in densely spaced features, such as a reduced separation distance between two adjacent fin-shaped active regions. Due to the reduced separation distance between two adjacent fin-shaped active regions, the dimension of the dummy fin may be reduced, increasing the possibility of unintentional electrical shorting between two adjacent source/drain features, thereby leading to a degraded device performance and/or device failures. Additionally, source/drain features that are disposed between two adjacent dummy fins may disadvantageously increase the parasitic capacitance of the FinFET or the MBC transistor.

The present disclosure provides semiconductor devices and methods for forming source/drain spacers extending along sidewall surfaces of to-be-formed source/drain features, thereby confining the lateral growth of the source/drain features. In an exemplary embodiment, a method includes providing fin-shaped active regions protruding from a substrate, forming cladding layers extending along sidewalls of the fin-shaped active regions, forming a dielectric feature over the substrate to fill space between two adjacent cladding layers, forming a gate structure over channel regions of the fin-shaped active regions and over a first portion of the cladding layers, performing an etching process to remove a second portion of the cladding layers not covered by the gate structure to form sidewall spacer trenches, forming a dielectric spacer in each of the sidewall spacer trenches, and after the forming of the dielectric spacers, forming source/drain features.

The various aspects of the present disclosure will now be described in more detail with reference to the figures. In that regard, FIG. 1 is a flowchart illustrating method 100 of forming a semiconductor device according to embodiments of the present disclosure. Method 100 is described below in conjunction with FIGS. 2A-17A, 2B-17B, 7C-9C, 13C,

17C, 8D-9D, 18-19, 20-21, and FIGS. 22A-22B, which are fragmentary cross-sectional views or top views of a workpiece 200 at different stages of fabrication according to embodiments of method 100. Method 100 is merely an example and is not intended to limit the present disclosure to what is explicitly illustrated therein. Additional steps may be provided before, during and after the method 100 and some steps described can be replaced, eliminated, or moved around for additional embodiments of the method. Not all steps are described herein in detail for reasons of simplicity. Because the workpiece 200 will be fabricated into a corresponding semiconductor device 200 upon conclusion of the fabrication processes, the workpiece 200 may be referred to as the semiconductor device 200 as the context requires. For avoidance of doubts, the X, Y and Z directions in the figures are perpendicular to one another and are used consistently throughout the figures. Throughout the present disclosure, like reference numerals denote like features unless otherwise excepted.

Referring to FIGS. 1, 2A and 2B, method 100 includes a block 102 where a workpiece 200 is received. FIG. 2A depicts a cross-sectional view of an exemplary workpiece 200 when viewed from the X direction, and FIG. 2B depicts a cross-sectional view of the workpiece 200 when viewed from the Y direction and taken along line B-B' shown in FIG. 2A. The workpiece 200 may be an intermediate structure fabricated during processing of an IC, or a portion thereof, that may comprise static random-access memory (SRAM) and/or other logic circuits, passive components such as resistors, capacitors, and inductors, and active components such as MBC transistors, FinFETs, metal-oxide semiconductor field effect transistors (MOSFETs), complementary metal-oxide semiconductor (CMOS) transistors, bipolar transistors, high voltage transistors, high frequency transistors, and/or other transistors. In the present embodiments, the workpiece 200 includes one or more MBC transistors. Additional features can be added to the workpiece 200, and some of the features described below can be replaced, modified, or eliminated in other embodiments of the workpiece 200.

As shown in FIGS. 2A and 2B, the workpiece 200 includes a substrate 202. In an embodiment, the substrate 202 is a bulk silicon substrate. The substrate 202 may include other semiconductor materials in various embodiments, such as germanium, silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, indium antimonide, SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, GaInAsP, or combinations thereof. In some alternative embodiments, the substrate 202 may be a semiconductor-on-insulator substrate, such as a silicon-on-insulator (SOI) substrate, a silicon germanium-on-insulator substrate, or a germanium-on-insulator (GeOI) substrate. The substrate 202 can include various doped regions configured according to design requirements of the semiconductor device 200. P-type doped regions may include p-type dopants, such as boron (B), boron difluoride (BF₂), other p-type dopant, or combinations thereof. N-type doped regions may include n-type dopants, such as phosphorus (P), arsenic (As), other n-type dopant, or combinations thereof. The various doped regions can be formed directly on and/or in the substrate 202, for example, providing a p-well structure, an n-well structure, or combinations thereof. An ion implantation process, a diffusion process, and/or other suitable doping process can be performed to form the various doped regions. Referring to FIG. 2A, the substrate 202 includes a first region 202N for formation of n-type MBC transistors and a second region 202P for formation of p-type MBC transis-

tors. The first region **202N** may include a p-type well and the second region **202P** may include an n-type well.

In embodiments represented in FIG. 2A, the workpiece **200** includes a fin-shaped active region **205a** formed over the second region **202P** of the substrate **202** and a fin-shaped active region **205b** formed over the first region **202N** of the substrate **202**. It is understood that the workpiece **200** may include any suitable number of fin-shaped active regions. The fin-shaped active regions **205a-205b** may be separately or collectively referred to as a fin-shaped active region **205** or fin-shaped active regions **205**. Each of the fin-shaped active regions **205** has a width **W1** along the Y direction. Each of the fin-shaped active regions **205** extends lengthwise along the X direction and is divided into channel regions **205C** (shown in FIG. 2B) and sources/drain regions **205S/D** (shown in FIG. 2B). The fin-shaped active region **205** may include a vertical stack of channel members in case of MBC transistors. In the embodiments represented in FIGS. 2A-2B, the semiconductor device **200** includes MBC transistors and each of the fin-shaped active regions **205** may be formed from a portion of the substrate **202** and a vertical stack **207** using a combination of lithography and etch steps. The vertical stack **207** includes a number of channel layers **208** interleaved by a number of sacrificial layers **206**. In embodiments represented in FIGS. 2A-2B, the vertical stack **207** includes four sacrificial layers **206** interleaved by three channel layers **208**. That is, the topmost sacrificial layer **206T** is formed on the topmost channel layer **208**. In some embodiments, a thickness of the topmost sacrificial layer **206T** may be different than a thickness of the rest of the sacrificial layers **206**. Each channel layer **208** may include a semiconductor material such as, silicon, germanium, silicon carbon, silicon germanium, or other suitable semiconductor materials, or combinations thereof, while each sacrificial layer **206** has a composition different from that of the channel layer **208**. The channel layers **208** and the sacrificial layers **206** may be epitaxially deposited on the substrate **202** using molecular beam epitaxy (MBE), vapor-phase epitaxy (VPE), ultra-high vacuum CVD (UHV-CVD), and/or other suitable epitaxial growth processes. In this depicted example, the channel layer **208** is formed of silicon (Si) and the sacrificial layer **206** is formed of silicon germanium (SiGe).

In some other embodiments, each of the fin-shaped active regions **205** may be a fin structure and the semiconductor device **200** may include FinFETs. The fin-shaped active regions **205** may include silicon (Si) or another elementary semiconductor, such as germanium (Ge); a compound semiconductor such as silicon carbide (SiC), gallium arsenide (GaAs), gallium phosphide (GaP), indium phosphide (InP), indium arsenide (InAs), and/or indium antimonide (InSb); an alloy semiconductor such as silicon germanium (SiGe), gallium arsenic phosphorus (GaAsP), aluminum indium arsenic (AlInAs), aluminum gallium arsenic (AlGaAs), indium gallium arsenic (InGaAs), gallium indium phosphorus (GaInP), and/or gallium indium arsenic phosphorus (GaInAsP); or combinations thereof.

The workpiece **200** also includes a hard mask layer **210** formed on the fin-shaped active region **205** to protect the fin-shaped active region **205** during subsequent etching processes. The hard mask layer **210** may include any suitable material and may be a single-layer structure or a multi-layer structure. In embodiments represented in FIG. 2A, the hard mask layer **210** includes a first hard mask layer **210a** formed on the fin-shaped active region **205** and a second hard mask layer **210b** formed on the first hard mask layer **210a**. A total thickness of the first hard mask layer

210a and the topmost sacrificial layer **206T** may determine a thickness of a helmet layer **220** (shown in FIG. 6A). In an embodiment, the first hard mask layer **210a** may include silicon nitride (SiN), the second hard mask layer **210b** may include silicon oxide (SiO₂).

Still referring to FIG. 2A, the fin-shaped active regions **205a-205b** are separated by an isolation structure **204**. The isolation structure **204** may include silicon oxide, tetraethylorthosilicate (TEOS), doped silicon oxide (e.g., borophosphosilicate glass (BPSG), fluoride-doped silicate glass (FSG), phosphosilicate glass (PSG), boron-doped silicate glass (BSG), etc.), a low-k dielectric material (having a dielectric constant less than that of silicon oxide, which is about 3.9), other suitable materials, or combinations thereof. The isolation structure **204** may include shallow trench isolation (STI) features. In one embodiment, the isolation structure **204** is formed by filling trenches that separate the fin-shaped active regions **205** with a dielectric material described above by any suitable method, such as CVD, flowable CVD (FCVD), spin-on-glass (SOG), other suitable methods, or combinations thereof. The dielectric material may subsequently be planarized by a chemical-mechanical planarization/polishing (CMP) process and selectively etched back to form the isolation structure **204**. After the etching back, a trench **212** is defined by the top surface of the isolation structure **204** and the sidewall surfaces of two adjacent fin-shaped active regions **205**. The trench **212** spans a width **S1** along the Y direction. That is, the separation distance between two adjacent fin-shaped active regions **205** may be referred to as **S1**. The isolation structure **204** may include a single-layer structure or a multi-layer structure. For example, the isolation structure **204** may include an oxide liner layer and a dielectric filler layer over the liner layer.

Referring to FIGS. 1, 3A and 3B, method **100** includes a block **104** where a cladding layer **214** is formed over the workpiece **200** to extend along sidewall surfaces of each fin-shaped active region **205**. As shown in FIG. 3A, the cladding layer **214** partially fills the trench **212**. In the present embodiments, the cladding layer **214** may have a composition that is substantially the same as that of the sacrificial layer **206**, such that they may be selectively removed by a common etching process. In the present embodiment, the cladding layer **214** is formed of SiGe. In an embodiment, the cladding layer **214** may be formed by an amorphous SiGe deposition process and is formed on the top surface of the hard mask layer **210**. In some embodiments, a thickness **T1** of the cladding layer **214** may determine a thickness of the source/drain spacer **230b** (shown in FIG. 13A).

Referring to FIGS. 1, 4A, and 4B, method **100** includes a block **106** where a dummy fin **216** is formed to substantially fill the trench **212**. In some embodiments, the dummy fin **216** may be a single-layer structure. In some other embodiments, the dummy fin **216** may include a multi-layer structure. For example, as shown in FIG. 4A, the dummy fin **216** includes a second dielectric layer **216b** embedded in a first dielectric layer **216a**. The formation of the dummy fin **216** may include conformally depositing the first dielectric layer **216a** over the workpiece **200** to partially fill the trench **212** by performing a deposition process such as a CVD process, a PVD process, an ALD process, or other suitable deposition process. The first dielectric layer **216a** may include silicon nitride, silicon carbon nitride (SiCN), silicon oxycarbonitride (SiOCN), or other suitable materials. In an embodiment, the first dielectric layer **216a** is formed of silicon carbon nitride. Subsequently, the second dielectric layer

216b is formed over the first dielectric layer **216a** to substantially fill the trench **212**. The second dielectric layer **216b** may be deposited over the workpiece **200** using CVD, SACVD, FCVD, ALD, spin-on coating, and/or other suitable process and may include silicon oxide, silicon carbide, FSG, or other suitable low-k dielectric materials. In an embodiment, the second dielectric layer **216b** is formed of silicon oxide. After the deposition of second dielectric layer **216b**, a planarization process, such as a CMP process, may be performed to planarize the workpiece **200** to remove excess materials and expose a top surface of the cladding layer **214**. The first dielectric layer **216a** and the second dielectric layer **216b** formed in the trench **212** may be collectively referred to as a dummy fin **216**. As depicted herein, the dummy fin **216** is separated from each sidewall of the fin-shaped active region **205** by the cladding layer **214**. The dummy fin **216** is configured to isolate adjacent fin-shaped active regions **205** and prevent the merge of source/drain features that will be formed in and over the adjacent fin-shaped active regions **205**. In some implementations, the dummy fin **216** may further operate with a helmet layer to isolate two adjacent gate stacks or divide a gate stack into two pieces.

Referring to FIGS. 1, 5A and 5B, method **100** includes a block **108** where the dummy fin **216** is selectively recessed to form a trench **218** between two adjacent fin-shaped active regions **205** using an etching process. In some embodiments shown in FIG. 5A, after performing the etching process, a top surface of the recessed dummy fin **216** is substantially coplanar with the top surface of the topmost channel layer **208**. The etching process may include any suitable process, such as a dry etching process, a wet etching process, other suitable processes, or combinations thereof.

Referring to FIGS. 1, 6A and 6B, method **100** includes a block **110** where a helmet layer **220** is formed in the trench **218**. The helmet layer **220** is deposited over the workpiece **200**, thereby filling the trench **218**. The helmet layer **220** may be a high-k dielectric layer and may include silicon nitride, aluminum oxide, aluminum nitride, aluminum oxynitride, zirconium oxide, zirconium nitride, zirconium aluminum oxide, hafnium oxide, other high-k material, or a suitable dielectric material. The helmet layer **220** may be deposited by a CVD process, an ALD process, a PVD process, and/or other suitable process. The workpiece **200** is then planarized using a CMP process to remove excess helmet layer **220** on the hard mask layer **210**. In embodiments represented in FIG. 6A, the CMP process stops after removing the second hard mask layer **210b**. In the present embodiment, the helmet layer **220** is configured to provide isolation for a to-be-formed gate stack **248** (shown in FIG. 17C) over the fin-shaped active region **205b** and a to-be-formed gate stack **250** (shown in FIG. 17C) over the fin-shaped active region **205a**. In some other examples, the helmet layer **220** may be configured to divide a gate stack into multiple portions. The helmet layer **220** may be referred to as a gate isolation feature or a gate cut feature. The dummy fin **216** and the helmet layer **220** may be collectively referred to as a hybrid fin.

Referring to FIGS. 1, 7A 7B, 7C (7A-7C), 8A, 8B, 8C, and 8D (8A-8D), method **100** includes a block **112** where a dummy gate structure **224** is formed over the channel regions **205C** of the fin-shaped active regions **205**. With reference to FIGS. 7A and 7B, the workpiece **200** is etched to selectively remove the topmost sacrificial layer **206T** and a top portion of the cladding layer **214** that extends along the sidewall surfaces of the topmost sacrificial layer **206T** to form a trench **222** without substantially etching the helmet

layer **220** or the topmost channel layer **208**. The trench **222** exposes the topmost channel layer **208**. In some implementations, the etching process may include a selective dry etching process. In some implementations, the etching process may include a selective wet etching process (e.g., selective to SiGe) that includes ammonium hydroxide (NH₄OH), hydrogen fluoride (HF), hydrogen peroxide (H₂O₂), or a combination thereof. After the etching process, top surfaces of the cladding layer **214** and the topmost channel layer **208** are substantially coplanar. FIG. 7C depicts a fragmentary top view of the workpiece **200** shown in FIGS. 7A-7B. As shown in FIG. 7C, the cladding layer **214** extends lengthwise along the sidewall surface of the fin-shaped active region **205** orientated in the X direction.

Subsequently, as shown in FIGS. 8A-8D, a dummy gate structure **224** is formed over the channel regions **205C** of the fin-shaped active regions **205**. FIG. 8C depicts a top view of the workpiece **200**. FIG. 8A depicts a cross-sectional view of the workpiece **200** taken along line A-A' shown in FIG. 8C. FIG. 8B depicts a cross-sectional view of the workpiece **200** taken along line B-B' shown in FIG. 8C. FIG. 8D depicts a cross-sectional view of the workpiece **200** taken along line D-D' shown in FIG. 8C. In this embodiment, a gate replacement process (or gate-last process) is adopted where the dummy gate structure **224** serves as a placeholder for a functional gate stack (e.g., gate stack **248** or gate stack **250**). Other processes and configurations are possible. While not explicitly shown, the dummy gate structure **224** may include a dummy dielectric layer and a dummy electrode disposed over the dummy dielectric layer. In some embodiments, the dummy dielectric layer may include silicon oxide and the dummy electrode may include polycrystalline silicon (polysilicon). As shown in FIG. 8B, the workpiece **200** also includes a gate-top hard mask layer **226** formed on the dummy gate structure **224**. In some embodiments, the gate-top hard mask layer **226** may be a multi-layer structure. The dummy gate structure **224** and the gate-top hard mask layer **226** may be formed by one or more deposition processes, photolithography process, and/or etching processes. In embodiments represented in FIG. 8C, the cladding layer **214** includes a first portion **214a** that is formed adjacent to the channel regions **205C** of the fin-shaped active regions **205** and disposed directly under the dummy gate structure **224**. The cladding layer **214** also includes a second portion **214b** that is formed adjacent to the source/drain regions **205S/D** of the fin-shaped active regions **205** and not covered by the dummy gate structure **224**.

Referring to FIGS. 1, 9A, 9B, 9C, and 9D, method **100** includes a block **114** where the second portion **214b** of the cladding layer **214** that is not covered by the dummy gate structure **224** is selectively removed by an etching process **227** to form trenches **228** (shown in FIGS. 9A and 9C) using the dummy gate structure **224** and the gate-top hard mask layer **226** as an etch mask. The first portion **214a** of the cladding layer **214** that is covered by the dummy gate structure **224** and the gate-top hard mask layer **226** is kept substantially intact. FIG. 9C depicts a top view of the workpiece **200**. FIG. 9A depicts a cross-sectional view of the workpiece **200** taken along line A-A' shown in FIG. 9C. FIG. 9B depicts a cross-sectional view of the workpiece **200** taken along line B-B' shown in FIG. 9C. FIG. 9D depicts a cross-sectional view of the workpiece **200** taken along line D-D' shown in FIG. 9C. As shown in FIG. 9A and FIG. 9C, after the removal of the second portion **214b** of the cladding layer **214**, the sidewall surfaces of the dummy fin **216** and the vertical stack **207** and a portion of the top surface of the STI structure **204** are exposed by the trenches **228**. The

trench 228 spans a width along the Y direction that is substantially equal to the thickness T1 (shown in FIG. 3A) of the cladding layer 214. The etching process 227 may be an anisotropic dry etching process that is configured to remove the second portion 214b of the cladding layer 214 without substantially etching the helmet layer 220, the dummy fin 216, the sacrificial layers 206 and the channel layers 208.

Referring to FIGS. 1, 10A, and 10B, method 100 includes a block 116 where a spacer layer 230 is deposited over the workpiece 200. After removing the second portion 214b of the cladding layer 214 and forming the trenches 228, the spacer layer 230 is deposited conformally over the workpiece 200, including over top surfaces of the gate-top hard mask layer 226, the helmet layer 220 and the fin-shaped active region 205 and sidewall surfaces of the dummy gate structure 224, the dummy fin 216 and the fin-shaped active regions 205. The term “conformally” may be used herein for ease of description of a layer having substantially uniform thickness over various regions. In embodiments represented in FIG. 10A, the spacer layer 230 substantially fills the trenches 228. That is, a deposition thickness T2 of the spacer layer 230 may be substantially equal to or greater than a half of the width T1 of the trench 228. Put differently, T2 is equal to or greater than $0.5 \cdot T1$. The spacer layer 230 may be deposited over the workpiece 200 using processes such as CVD, atomic layer deposition (ALD), PVD, or other suitable process.

Dielectric materials for the spacer layer 230 may be selected to allow selective removal of the dummy gate structure 224 without substantially damaging the spacer layer 230, selective recess of the spacer layer 230 without substantially damaging the helmet layer 220 and the fin-shaped active region 205, and selective removal of the source/drain regions 205S/D of the fin-shaped active region 205 without substantially etching the spacer layer 230. Suitable dielectric materials may include silicon nitride, silicon oxycarbonitride, silicon carbonitride, silicon oxide, silicon oxycarbide, silicon carbide, silicon oxynitride, other low-k dielectric materials, and/or combinations thereof. In some embodiments, since the spacer layer 230 is in direct contact with the fin-shaped active regions 205, a material of the spacer layer 230 may be selected such that the to-be-formed source/drain spacers 230b (shown in FIG. 11A) may introduce strains to source/drain features (e.g., source/drain features 236N and 236P shown in FIG. 17A) that would be formed in and over the fin-shaped active regions 205, thereby increasing the carrier mobility of the workpiece 200. Exemplary strained materials may include nitride-based dielectric materials such as silicon nitride.

Referring to FIGS. 1, 11A, and 11B, method 100 includes a block 118 where the spacer layer 230 is etched back to form source/drain spacers 230b in the trenches 228 and gate spacers 230a along sidewall surfaces of the dummy gate structure 224. An anisotropic etching process may be performed to selectively remove portions of the spacer layer 230 that are not extending along sidewall surfaces of the fin-shaped active regions 205, the dummy fin 216, and the dummy gate structure 224, thereby forming source/drain spacers 230b in the trenches 228 and gate spacers 230a along sidewall surfaces of the dummy gate structure 224. The anisotropic etching process may include an anisotropic dry etching process. In the present embodiment, after the anisotropic etching process, a top surface of the source/drain spacer 230b is substantially coplanar with the top surface of the topmost channel layer 208. As shown in FIGS. 11A-11B, the source/drain spacer 230b is in direct contact with side-

wall surfaces of both the dummy fin 216 and the fin-shaped active region 205. That is, the source/drain spacer 230b is laterally sandwiched between the fin-shaped active region 205 and the dummy fin 216 along the Y direction. The regions of the fin-shaped active regions 205 underlying the dummy gate structure 224 may be referred to as channel regions 205C. Each of the channel regions 205C in the fin-shaped active region 205 is sandwiched between two source/drain regions 205S/D.

Referring to FIGS. 1, 12A, and 12B, method 100 includes a block 120 where source/drain regions 205S/D of the fin-shaped active regions 205 are selectively recessed to form source/drain trenches 232. With the gate-top hard mask layer 226, the gate spacers 230a, the source/drain spacers 230b, and the helmet layer 220 serving as an etch mask, the source/drain regions 205S/D of the fin-shaped active regions 205 are etched to form source/drain trenches 232 without substantially etching the helmet layer 220, the source/drain spacers 230b, the gate-top hard mask layer 226, and the gate spacers 230a. Source/drain trenches 232 may not only extend through the vertical stack 207, but also partially extend into the substrate 202. Due to the formation of the source/drain spacers 230b, a width of the source/drain trench 232 along the Y direction may be substantially equal to the width W1 of the fin-shaped active region 205.

Still referring to FIGS. 1, 12A, and 12B, method 100 includes a block 122 where inner spacer features 234 are formed. After forming the source/drain trenches 232, the sacrificial layers 206 exposed in the source/drain trenches 232 are selectively and partially recessed to form inner spacer recesses (filled by inner spacer features 234), while the exposed channel layers 208 are substantially unetched. In some embodiments, this selective recess may include a selective isotropic etching process (e.g., a selective dry etching process or a selective wet etching process), and the extent at which the sacrificial layers 206 is recessed is controlled by duration of the etching process. After the formation of the inner spacer recesses, an inner spacer material layer is then conformally deposited using CVD or ALD over the workpiece 200, including over and into the inner spacer recesses. The inner spacer material may include silicon nitride, silicon oxycarbonitride, silicon carbonitride, silicon oxide, silicon oxycarbide, silicon carbide, or silicon oxynitride. The inner spacer material layer is then etched back to form inner spacer features 234, as illustrated in FIG. 12B. In some embodiments, a composition of the inner spacer features 234 is different than a composition of the gate spacers 230a and the source/drain spacers 230b such that the etching back of the inner spacer material layer does not substantially etch the gate spacers 230a and the source/drain spacers 230b.

Referring to FIGS. 1, 13A, 13B, and 13C, method 100 includes a block 124 where N-type source/drain features 236N are epitaxially formed in the corresponding source/drain trenches 232 over the first region 202N and P-type source/drain features 236P are epitaxially formed in the corresponding source/drain trenches 232 over the second region 202P by using one or more epitaxial processes, such as VPE, UHV-CVD, MBE, and/or other suitable processes. The source/drain features 236N and/or 236P are therefore coupled to the corresponding channel layers 208 in the channel regions 205C of the respective fin-shaped active regions 205. As exemplary shown in FIG. 13A, two adjacent source/drain features (e.g., the N-type source/drain feature 236N and the P-type source/drain feature 236P) are separated by the combination of the dummy fin 216 and the source/drain spacers 230b. That is, by forming the source/

drain spacers **230b**, a distance **W2** between the N-type source/drain feature **236N** and the P-type source/drain feature **236P** may be increased compared to embodiments where the source/drain features are separated only by the dummy fin **216**. In an embodiment, the distance **W2** is substantially equal to the separation distance **S1** (shown in FIG. 2A) between two adjacent fin-shaped active regions **205** and is greater than a width of the dummy fin **216** along the Y direction. In some embodiments, the distance **W2** may be between about 10 nm and about 45 nm. By increasing the distance **W2** between the N-type source/drain feature **236N** and the P-type source/drain feature **236P**, the possibility of the merge of the N-type source/drain feature **236N** with the P-type source/drain feature **236P** is reduced, and the possibility of the unintentional electrical shorting may be also reduced. In some embodiments, since the lateral growth of the epitaxial source/drain features **236N/236P** are confined by the source/drain spacers **230b** rather than the dummy fin **216**, the width of the dummy fin **216** may be reduced and thus the separation distance **S1** between two adjacent fin-shaped active regions **205** may be reduced, thereby increasing the density of fin-shaped active regions **205** on the substrate **202** and increasing the density of transistors.

In embodiments represented in FIG. 13A and FIG. 13B, the N-type source/drain feature **236N** and the P-type source/drain feature **236P** each includes a first semiconductor layer **236a** formed over top surfaces of the substrate **202** exposed in the source/drain trenches **232** by using an epitaxial process. In embodiments represented in FIG. 13A, the first semiconductor layer **236a** is in direct contact with a sidewall surface of the source/drain spacer **230b**. The first semiconductor layer **236a** may be undoped or not intentionally doped and may be configured to reduce leakage through the substrate **202**. In some embodiments, the first semiconductor layer **236a** may include undoped silicon (Si), undoped germanium (Ge), undoped silicon germanium (SiGe), or other suitable materials.

The N-type source/drain feature **236N** and the P-type source/drain feature **236P** each also includes a corresponding second semiconductor layer **236b** and second semiconductor layer **236b'** formed over top surfaces of the first semiconductor layer **236a**, respectively. Compositions of the second semiconductor layers **236b** and **236b'** are different than a composition of the first semiconductor layer **236a**. More specifically, the second semiconductor layers **236b** of the N-type source/drain feature **236N** may include arsenic-doped silicon (Si:As), phosphorus-doped silicon (Si:P), arsenic-doped silicon carbide (SiC:As), phosphorus-doped silicon carbide (SiC:P) or other suitable materials, and have a dopant concentration greater than that of the undoped first semiconductor layer **236a**. The second semiconductor layers **236b'** of the P-type source/drain feature **236P** may include boron-doped silicon germanium (SiGe:B) or other suitable materials, and have a dopant concentration greater than that of the undoped first semiconductor layer **236a**. In some embodiments, the second semiconductor layers **236b** and **236b'** may be selectively grown from semiconductor surfaces exposed in the source/drain trenches **232** by using an epitaxial process. The epitaxial process may use gaseous and/or liquid precursors, which interact with the composition of the first semiconductor layer **236a**. In some implementations, the second semiconductor layers **236b** and **236b'** each may include a bottom segment on the first semiconductor layer **236a** and multiple sidewall segments on the sidewall surfaces of the channel layers **208**, thereby partially filling the source/drain trenches **232**. In embodiments represented in FIG. 13A, the bottom segments of the second

semiconductor layers **236b** and **236b'** are in direct contact with sidewall surfaces of the source/drain spacers **230b**.

The N-type source/drain feature **236N** and the P-type source/drain feature **236P** may each include a corresponding third semiconductor layer **236c** and third semiconductor layer **236c'** formed over the second semiconductor layers **236b** and the second semiconductor layers **236b'**, respectively, to fill the corresponding source/drain trenches **232**. The third semiconductor layers **236c-236c'** may be formed in the source/drain trenches **232** by using an epitaxial process, such as VPE, UHV-CVD, MBE, and/or other suitable processes. A material of the third semiconductor layer **236c** may be the same as or different than a material of the second semiconductor layers **236b**, and a material of the third semiconductor layer **236c'** may be the same as or different than a material of the second semiconductor layers **236b'**, a dopant concentration of the third semiconductor layer **236c** is greater than that of the second semiconductor layers **236b**, and a dopant concentration of the third semiconductor layer **236c'** is greater than that of the second semiconductor layers **236b'**. In some embodiments, the third semiconductor layer **236c** and the third semiconductor layer **236c'** are heavily doped semiconductor layers. The third semiconductor layer **236c** of the N-type source/drain feature **236N** may include arsenic-doped silicon (Si:As), phosphorus-doped silicon (Si:P), arsenic-doped silicon carbide (SiC:As), phosphorus-doped silicon carbide (SiC:P) or other suitable materials. The third semiconductor layer **236c'** of the P-type source/drain feature **236P** may include boron-doped silicon germanium (SiGe:B) or other suitable materials. In embodiments represented in FIG. 13A, the third semiconductor layer **236c** and **236c'** are in direct contact with sidewall surfaces of the source/drain spacers **230b**. That is, in embodiments represented in FIG. 13A, the first, second, and third semiconductor layers of the source/drain feature **236N/236P** are all in direct contact with the source/drain spacers **230b**.

In embodiments represented in FIG. 13A, a top surface of the source/drain feature (e.g., the N-type source/drain feature **236N** and/or the P-type source/drain feature **236P**) is substantially coplanar with a top surface of the source/drain spacers **230b**. It is further noted that, as shown in FIG. 13A and FIG. 13C, the lateral growth of the source/drain feature **236N/236P** along the Y direction is confined by the source/drain spacers **230b**. As such, the source/drain feature **236N/236P** has a uniform width substantially equal to the width **W1** of the fin-shaped active region **205** bottom to top and when viewed from the X direction, the sidewalls of the source/drain feature **236N/236P** are substantially vertical. In some other implementations, a top surface of the source/drain feature **236N/236P** may be above a top surface of the source/drain spacers **230b** and below a top surface of the helmet layer **220**. A distance **H** between the top surface of the source/drain feature **236N/236P** and a top surface of the helmet layer **220** may be less than about 20 nm.

Referring to FIGS. 1, 14A, 14B, 15A, and 15B, method **100** includes a block **126** where the dummy gate structure **224**, the first portion **214a** of the cladding layer **214**, and the sacrificial layers **206** are selectively removed. As shown in FIGS. 14A-14B, a contact etch stop layer (CESL) **240** and an interlayer dielectric (ILD) layer **242** are deposited over the workpiece **200**. The CESL **240** may include silicon nitride, silicon oxynitride, and/or other suitable materials and may be formed by ALD, plasma-enhanced chemical vapor deposition (PECVD) process and/or other suitable deposition or oxidation processes. As shown in FIGS. 14A-14B, the CESL **240** may be deposited on top surfaces of the

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source/drain features **236N/236P**, the source/drain spacers **230b** and the helmet layer **220**, and sidewall surfaces of the gate spacers **230a** and the helmet layer **220**. The ILD layer **242** may be deposited by a PECVD process, FCVD, or other suitable deposition technique over the workpiece **200** after the deposition of the CESL **240**. The ILD layer **242** may include materials similar to that of the isolation structures **204**. A planarization process may be then performed to remove excess materials such as removing the gate-top hard mask layer **226** on the dummy gate structure **224** and portions of the CESL **240** and the ILD layer **242** over the gate-top hard mask layer **226**.

Subsequently, referring to FIGS. **15A-15B**, the dummy gate structure **224** is selectively removed by an etching process. The removal of the dummy gate structure **224** forms gate trenches **244** over the channel regions **205C**. Although not shown, it is understood that the gate trenches **244** expose the first portion **214a** of the cladding layer **214** that was covered by the dummy gate structure **224**. The etching process for removing the dummy gate structure **224** may include any suitable process, such as a dry etching process, a wet etching process, or combinations thereof, and is configured to selectively remove the dummy gate structure **224** without substantially etching the channel layers **208**, the gate spacer **230a**, the helmet layer **220**, the dummy fin **216**, the CESL **240**, and the ILD layer **242**.

After the selective removal of the dummy gate structure **224**, without substantially removing the channel layers **208**, one or more etching processes may be performed to selectively remove the first portion **214a** of the cladding layer **214** and the sacrificial layers **206** to release the channel layers **208** as channel members **208**. Since the composition of the cladding layer **214** is the same as the composition of the sacrificial layers **206**, the first portion **214a** of the cladding layer **214** and the sacrificial layers **206** may be removed by a common etching process. In one example, the etching process for removing the sacrificial layers **206** may be a wet etching process that employs an oxidant such as ammonium hydroxide (NH_4OH), ozone (O_3), nitric acid (HNO_3), hydrogen peroxide (H_2O_2), other suitable oxidants, and a fluorine-based etchant such as hydrofluoric acid (HF), ammonium fluoride (NH_4F), other suitable etchants, or combinations thereof. The removal of the sacrificial layers **206** forms a number of openings **246**. Although not shown, it is understood that the removal of the first portion **214a** of the cladding layer **214** would form a trench disposed between the stack of channel members **208** and the dummy fin **216**.

Referring to FIGS. **1**, **16A**, and **16B**, method **100** includes a block **128** where a gate stack **248** is formed over the first region **202N** of the substrate **202** and a gate stack **250** (shown in FIG. **17C**) is formed over the second region **202P** of the substrate **202** to fill the gate trenches **244** and the openings **246** and wrap around each of the channel members **208**. The gate stacks **248** and **250** each may include an interfacial layer (not separately labeled), a gate dielectric layer (not separately labeled) over the interfacial layer, and a corresponding gate electrode layer (not separately labeled) over the gate dielectric layer. In some embodiments, the interfacial layer may include silicon oxide formed by thermal oxidization. The gate dielectric layer is then deposited over the interfacial layer using ALD, CVD, and/or other suitable methods. The gate dielectric layer may include high-k dielectric materials. As used herein, high-k dielectric materials include dielectric materials having a high dielectric constant, for example, greater than that of thermal silicon oxide (~ 3.9). In one embodiment, the gate dielectric layer may include hafnium oxide. Alternatively, the gate

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dielectric layer may include other high-k dielectrics, such as titanium oxide (TiO_2), hafnium zirconium oxide (HfZrO), tantalum oxide (Ta_2O_5), hafnium silicon oxide (HfSiO_4), zirconium oxide, zirconium silicon oxide (ZrSiO_2), lanthanum oxide (La_2O_3), aluminum oxide (Al_2O_3), yttrium oxide (Y_2O_3), SrTiO_3 (STO), BaTiO_3 (BTO), BaZrO , hafnium lanthanum oxide (HfLaO), lanthanum silicon oxide (La-SiO), aluminum silicon oxide (AlSiO), hafnium tantalum oxide (HfTaO), hafnium titanium oxide (HfTiO), (Ba,Sr) TiO_3 (BST), silicon nitride (SiN), silicon oxynitride (SiON), combinations thereof, or other suitable materials.

A gate electrode layer is deposited over the gate dielectric layer. The gate electrode layer may be a multi-layer structure that includes at least one work function layer and a metal fill layer. By way of example, the at least one work function layer may include titanium nitride (TiN), titanium aluminum (TiAl), titanium aluminum nitride (TiAlN), tantalum nitride (TaN), tantalum aluminum (TaAl), tantalum aluminum nitride (TaAlN), tantalum aluminum carbide (TaAlC), tantalum carbonitride (TaCN), or tantalum carbide (TaC). The metal fill layer may include aluminum (Al), tungsten (W), nickel (Ni), titanium (Ti), ruthenium (Ru), cobalt (Co), platinum (Pt), tantalum silicon nitride (TaSiN), copper (Cu), other refractory metals, or other suitable metal materials or a combination thereof. It is understood that the gate stack **248** and the gate stack **250** may include different work function layers. In various embodiments, the gate electrode layer may be formed by ALD, PVD, CVD, e-beam evaporation, or other suitable process. In various embodiments, a planarization process, such as a CMP process, may be performed to remove excess portions of those materials to provide a substantially planar top surface of the gate stack **248/250**. The planarization process may stop when it reaches or before reaching the top surface of the helmet layer **220**. In some embodiments, due to the formation of the source/drain spacers **230b**, the lateral growth of the source/drain features **236N/236P** along the Y direction is confined, leading to a reduced overlapping area between the source/drain feature **236N/236P** and the gate stack **248/250**. In this regard, the reduction in the overlapping area lowers the parasitic gate-drain capacitance (C_{gd}) of the MBC transistor, thereby reducing RC delay and increasing processing speed the MBC transistor when applied in alternating current (AC) settings.

Referring to FIGS. **1**, **17A**, **17B**, and **17C**, method **100** includes a block **130** where further processes may be performed to complete the fabrication of the semiconductor device **200**. Such further processes may include, for example, as shown in FIGS. **17A-17B**, performing an etching process to selectively recess the gate stack (e.g., the gate stack **248**) until a portion of the helmet layer **220** protrudes from a top surface of the recessed gate stack **248**. The etching process may include any suitable process, such as a dry etching process, a wet etching process, an RIE, or combinations thereof, configured to selectively remove the gate stack **248** without substantially etching the helmet layer **220**, the gate spacers **230a**, the CESL **240**, and the ILD layer **242**. The resulting height of the recessed gate stack **248** may be controlled by the duration of the etching process. In the present embodiments, after the selective recessing of the gate stack **248** and the gate stack **250**, a top surface of the recessed gate stack **248** and a top surface of the recessed gate stack **250** are below the top surface of the helmet layer **220**. As such, the combination of the dummy fin **216** and the helmet layer **220** electrically and physically isolates the gate stack **250** from the gate stack **248**, such that gate terminals of the transistor formed over the first region **202N** and the

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transistor formed over the formed over the second region **202P** may be separately controlled.

Such further processes may also include, for example, performing another etching process to selectively recess the gate spacers **230a**. The selective recessing of the gate stack **248** and selective recess of the gate spacers **230a** form a cap trench. Such further processes may also include, for example, as shown in FIGS. **17B** and **17C**, depositing a dielectric layer **252** over the workpiece **200**, thereby filling the cap trench. In the present embodiments, the dielectric layer **252** is configured to provide self-alignment capability and etching selectivity during subsequent fabrication processes including, for example, patterning the ILD layer **242** to form source/drain contact openings over the epitaxial source/drain features **236N/236P**. Accordingly, in the present embodiments, the dielectric layer **252** has a composition different from that of the ILD layer **242**. In some embodiments, the dielectric layer **252** may include SiN, SiCN, SiOC, SiON, SiOCN, other suitable materials, or combinations thereof and may be deposited by any suitable method, such as ALD, CVD, PVD, other suitable methods, or combinations thereof. Subsequently, portions of the dielectric layer **252** formed over the ILD layer **242** may be removed by one or more CMP process, thereby providing the workpiece **200** a planar top surface. As illustrated in FIGS. **17A** and **17C**, a portion of the helmet layer **220** is embedded in the dielectric layer **252**.

Such further processes may also include forming a multi-layer interconnect (MLI) structure (not depicted) thereover. The MLI structure may include various interconnect features, such as vias and conductive lines, disposed in dielectric layers, such as etch stop layers (ESLs) and ILD layers. In some embodiments, the vias are vertical interconnect features configured to interconnect a device-level contact, such as an S/D contact (not depicted) or a gate contact (not depicted), with a conductive line or interconnect different conductive lines, which are horizontal interconnect features. The ESLs and the ILD layers of the MLI structure may have substantially same compositions as those discussed above with respect to the CESL **240** and the ILD layer **242**, respectively. The vias and the conductive lines may each include any suitable conductive material, such as Co, W, Ru, Cu, Al, Ti, Ni, Au, Pt, Pd, a metal silicide, other suitable conductive materials, or combinations thereof, and be formed by a series of patterning and deposition processes. Additionally, each via and conductive line may additionally include a barrier layer that comprises TiN and/or TaN.

FIGS. **18** and **19** depict a first alternative embodiment. As represented in FIG. **18**, one or more airgaps **231** (i.e., seams or voids) may be formed during the deposition of the spacer layer **230**. In the present embodiment, the airgap **231** is formed in a portion of the spacer layer **230** disposed laterally between the fin-shaped active region **205** and dummy fin **216**. Upon conclusion of the fabrication process, as presented by FIG. **19**, the semiconductor device **200** includes the airgap **231** disposed between the source/drain feature **236P/236N** and the dummy fin **216**. The formation of the airgaps **231** may further reduce the parasitic capacitance of the semiconductor device **200**.

In the embodiments described above, the fin-shaped active region **205** is kept substantially intact after the performing of the etching process **227** (shown in FIG. **9A**). Since the channel layers **208** and the sacrificial layers **206** in the source/drain regions **205S/D** would be removed by a subsequent etching process to form source/drain trenches **232** (shown in FIG. **12A**), etchants and/or other parameters (e.g., pressure, gas flow, temperature) used in the etching

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process **227** may be configured to adjust the etchant selectivity between the cladding layer **214** and the channel layer **208**, thereby adjusting the height of the fin-shaped active region **205** after the etching process **227** and adjusting a height of the source/drain spacers **230b**.

FIGS. **20-21** depict an alternative embodiment where a height of the source/drain spacer **230b** may be changed. In some embodiments, during the formation of the source/drain trenches **232**, the source/drain spacer **230b** may be recessed. The recessed source/drain spacer **230b** may be referred to as source/drain spacer **230b'**. A top surface of the source/drain spacer **230b'** is below the top surface of the dummy fin **216** and the height of the source/drain spacer **230b'** is smaller than a height of the dummy fin **216**. After forming the source/drain spacer **230b'**, operations in blocks **122-130** of method **100** may be performed, thereby finishing the fabrication of the workpiece **200**. FIG. **21** depicts a cross-sectional view of the workpiece **200** after forming source/drain features **236P'** and **236N'**. The source/drain feature **236P'** is in a way similar to the source/drain feature **236P** and the source/drain feature **236N'** is in a way similar to the source/drain feature **236N** except that the profiles of the source/drain features **236P'** and **236N'** are different than those of the source/drain features **236P** and **236N** when viewed from the X direction. More specifically, in embodiments described with reference to FIGS. **13A-13C**, the source/drain features **236P** and **236N** have a substantially uniform width **W1** bottom to up and the width **W1** is substantially equal to the width of the fin-shaped active region **205** along the Y direction. In embodiments represented in FIG. **21**, due to the reduced height of the source/drain spacers **230b'**, with a lower portion of the source/drain features **236P'** and **236N'** being confined laterally along the Y direction, an upper portion of the source/drain features **236P'** and **236N'** is allowed to extend laterally along the Y direction, overhangs the lower portion of the source/drain features **236P'** and **236N'**, and disposed directly over the source/drain spacers **230b'**. That is, a width of the upper portion of the source/drain features **236P'** and **236N'** may be greater than the width **W1** of the lower portion of the source/drain features **236P'** and **236N'**.

In embodiments described above with reference to the FIGS. **10A-10B**, the spacer layer **230** is a single-layer structure. In some other embodiments, the spacer layer **230** may be a multi-layer structure that includes two or more dielectric layers. For example, the formation of the spacer layer **230** may include forming a first dielectric layer (e.g., a first dielectric layer **2301** shown in FIG. **22A**) conformally over the workpiece **200** and forming a second dielectric layer (e.g., second dielectric layer **2302** shown in FIG. **22A**) deposited conformally over the first dielectric layer. In some implementations, a dielectric constant of the first dielectric layer may be greater than that of the second dielectric layer, and the first dielectric layer is more etch resistant than the second dielectric layer. In some embodiments, the first dielectric layer may include silicon carbonitride, silicon nitride, zirconium oxide, aluminum oxide, or a suitable dielectric material, the second dielectric layer may include silicon oxide, silicon oxycarbide, or a suitable low-k dielectric material. In an embodiment, the spacer layer **230** includes a silicon oxide layer formed on a silicon nitride layer. After forming the multi-layer spacer layer **230**, operations in blocks **118-130** may be performed, thereby finishing the fabrication of the workpiece **200**. FIGS. **22A** and **22B** depict cross-sectional views of the workpiece **200** that includes gate spacers **230a''** and source/drain spacers **230b''**. The gate spacers **230a''** include the second dielectric layer

2302 formed on the first dielectric layer **2301** and the source/drain spacers **230b**" include the second dielectric layer **2302** embedded in the first dielectric layer **2301**. That is, although the gate spacers **230a**" and the source/drain spacers **230b**" are formed by common deposition processes and a common etching process, the resulted structures of the gate spacers **230a**" and the source/drain spacers **230b**" are different. It is understood that the structure of the source/drain spacers **230b**" may depend on the width T1 (shown in FIG. 9A) of the trenches **228** and the deposition thicknesses of the first and second dielectric layers **2301** and **2302**.

Although not intended to be limiting, one or more embodiments of the present disclosure provide many benefits. For example, the present disclosure includes forming source/drain spacers defining a source/drain trench and then forming a source/drain feature in the source/drain trench. As such, the lateral growth of the source/drain feature may be confined by the source/drain spacers, reducing an overlapping area between the source/drain feature and the gate stack of a transistor and thus reducing a parasitic capacitance. The reduced parasitic capacitance may thus lead to an increased device speed. The lateral confinement of the source/drain features may also advantageously increase the distance between two adjacent source/drain features, thereby reducing the possibility of unintentional electrical shorting between two adjacent source/drain features. In some embodiments, due to the formation of the source/drain spacers, the dummy fin may be configured to have a smaller width, thereby reducing the spacing of two adjacent active regions and increasing the device density. In some implementations, the source/drain spacers may be tensile strained dielectric spacers such that the source/drain features may be strained, and the carrier mobility may be enhanced.

The present disclosure provides for many different embodiments. Semiconductor devices and methods of fabrication thereof are disclosed herein. In one exemplary aspect, the present disclosure is directed to a method. The method includes providing a workpiece including an active region protruding from a substrate and comprising a channel region disposed adjacent to a source/drain region, a cladding layer extending along a sidewall surface of the active region, and an isolation structure spaced apart from the active region by the cladding layer. The method also includes forming a dummy gate structure over the channel region of the active region and a first portion of the cladding layer, after the forming of the dummy gate structure, selectively removing a second portion of the cladding layer not covered by the dummy gate structure to form a trench between the isolation structure and the active region, forming a dielectric spacer the trench, recessing the source/drain region of the active region to form a source/drain opening without substantially etching the dielectric spacer, forming a source/drain feature to fill the source/drain opening, and replacing the dummy gate structure with a gate stack.

In some embodiments, the forming of the dielectric spacer may include depositing a dielectric layer over the workpiece and etching back the dielectric layer to form the dielectric spacer in the trench. In some embodiments, the dielectric layer may be a conformal dielectric layer, and the etching back of the dielectric layer may also form a gate spacer extending along a sidewall surface of the dummy gate structure. In some embodiments, the depositing of the dielectric layer may further form an air gap within the dielectric layer, and the dielectric spacer may include the air gap. In some embodiments, the depositing of the dielectric layer may include conformally depositing a first dielectric layer over the workpiece and conformally depositing a

second dielectric layer over the first dielectric layer. A dielectric constant of the first dielectric layer may be greater than a dielectric constant of the second dielectric layer. In some embodiments, the active region may include a vertical stack of alternating channel layers and sacrificial layers, and a composition of the cladding layer may be the same as a composition of the sacrificial layers. In some embodiments, the replacing of the dummy gate structure with the gate stack may include selectively removing the dummy gate structure to form a first opening, selectively removing the sacrificial layers and the portion of the cladding layer to form second openings and forming the gate stack in the first and second openings. In some embodiments, the method may also include, after the forming of the source/drain opening, selectively recessing the sacrificial layers to form inner spacer recesses, depositing a dielectric material over the workpiece to fill the inner spacer recesses, and etching back the dielectric material to form inner spacer features in the inner spacer recesses. In some embodiments, the selectively removing of the second portion of the cladding layer may include performing an anisotropic dry etching process. In some embodiments, the isolation structure may include a lower portion having a first composition and an upper portion having a second composition different from the first composition, a top surface of the lower portion may be substantially coplanar with a top surface of the dielectric spacer and a top surface of the active region.

In another exemplary aspect, the present disclosure is directed to a method. The method includes providing a first fin-shaped active region and a second fin-shaped active region each protruding from a substrate and oriented lengthwise in a first direction, forming cladding layers extending along sidewalls of the first and second fin-shaped active regions and oriented lengthwise in the first direction, forming a dielectric feature over the substrate to fill space between two adjacent cladding layers, forming a gate structure over channel regions of the first and second fin-shaped active regions, the gate structure orientated lengthwise in a second direction substantially perpendicular to the first direction and covering a first portion of the cladding layers, performing an etching process to remove a second portion of the cladding layers not covered by the gate structure to form sidewall spacer trenches, forming a dielectric spacer in each of the sidewall spacer trenches, and after the forming of the dielectric spacers, forming source/drain features in source/drain regions of the first and second fin-shaped active regions.

In some embodiments, the forming of the dielectric spacers may include depositing a conformal dielectric layer over the substrate, and selectively recessing the conformal dielectric layer to form the dielectric spacer in each of the sidewall spacer trenches. The selectively recessing of the conformal dielectric layer may further form gate spacers extending along sidewalls of the gate structure. In some embodiments, the forming of the source/drain features may include selectively removing the source/drain regions of the first and second fin-shaped active regions to form source/drain openings, epitaxially forming a first semiconductor layer in the source/drain openings, epitaxially forming a second semiconductor layer over the first semiconductor layer, and epitaxially forming a third semiconductor layer over the second semiconductor layer to fill the source/drain openings, a dopant concentration of the third semiconductor layer may be greater than a dopant concentration of the second semiconductor layer. In some embodiments, the first semiconductor layer, the second semiconductor layer, and the third semiconductor layer may be in direct contact with the

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dielectric spacer. In some embodiments, the method may also include, after the forming of the source/drain features, selectively removing the gate structure to form a gate trench, the gate trench exposing the portion of the cladding layers, selectively removing the first portion of the cladding layers, and forming a gate stack over the substrate to fill the gate trench. In some embodiments, the performing of the etching process selectively may remove the second portion of the cladding layers without substantially etching the first and second fin-shaped active regions, and a height of the dielectric spacer may be substantially equal to a height of the first and second fin-shaped active regions. In some embodiments, a top surface of the dielectric spacer may be below a top surface of the source/drain features.

In yet another exemplary aspect, the present disclosure is directed to a semiconductor device. The semiconductor device includes a plurality of channel members disposed over a substrate and coupled to a source/drain feature along a first direction, a gate structure wrapping around each channel member of the plurality of channel members, and a dielectric spacer disposed adjacent to the source/drain feature along a second direction substantially perpendicular to the first direction, the source/drain feature includes at least two semiconductor layers, and each of the at least two semiconductor layers is in direct contact with the dielectric spacer.

In some embodiments, the semiconductor device may also include gate spacers extending along sidewalls of the gate structure, and a composition of the dielectric spacer may be the same as a composition of the gate spacers. In some embodiments, the semiconductor device may also include a dummy fin disposed adjacent to the dielectric spacer and a helmet layer disposed on the dummy fin without being disposed on the dielectric spacer, the dielectric spacer may be disposed between the dummy fin and the source/drain feature along the second direction and a composition of the helmet layer may be different from a composition of the dummy fin.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A method, comprising:

providing a workpiece comprising:

an active region protruding from a substrate and comprising a channel region disposed adjacent to a source/drain region,

a cladding layer extending along a sidewall surface of the active region, and

an isolation structure spaced apart from the active region by the cladding layer;

forming a dummy gate structure over the channel region of the active region, wherein the dummy gate structure covers a first portion of the cladding layer;

after the forming of the dummy gate structure, selectively removing a second portion of the cladding layer not

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covered by the dummy gate structure, thereby forming a trench between the isolation structure and the active region;

forming a dielectric spacer in the trench

recessing the source/drain region of the active region to form a source/drain opening without substantially etching the dielectric spacer;

forming a source/drain feature to fill the source/drain opening; and

after the forming of the source/drain feature, replacing the dummy gate structure with a gate stack.

2. The method of claim 1, wherein the forming of the dielectric spacer comprises:

depositing a dielectric layer; and

etching back the dielectric layer to form the dielectric spacer in the trench.

3. The method of claim 2,

wherein the etching back of the dielectric layer further forms a gate spacer extending along a sidewall surface of the dummy gate structure.

4. The method of claim 2, wherein the depositing of the dielectric layer further forms an air gap within the dielectric layer, and the dielectric spacer comprises the air gap.

5. The method of claim 1,

wherein the active region comprises a vertical stack of alternating channel layers and sacrificial layers, and wherein a composition of the cladding layer is the same as a composition of the sacrificial layers.

6. The method of claim 5, wherein the replacing of the dummy gate structure with the gate stack comprises:

selectively removing the dummy gate structure;

selectively removing the sacrificial layers and the first portion of the cladding layer; and

forming the gate stack wrapping around the channel layers.

7. The method of claim 6, further comprising:

after the forming of the source/drain opening, selectively recessing the sacrificial layers to form inner spacer recesses;

depositing a dielectric material over the workpiece to fill the inner spacer recesses; and

etching back the dielectric material to form inner spacer features in the inner spacer recesses.

8. The method of claim 1, wherein the selectively removing of the second portion of the cladding layer comprises performing an anisotropic dry etching process.

9. The method of claim 1, wherein the isolation structure comprises a lower portion having a first composition and an upper portion having a second composition different from the first composition,

wherein a top surface of the lower portion is substantially coplanar with a top surface of the dielectric spacer and a top surface of the active region.

10. A method, comprising:

providing a first fin-shaped active region and a second fin-shaped active region each protruding from a substrate and oriented lengthwise in a first direction;

forming cladding layers extending along sidewalls of the first and second fin-shaped active regions and oriented lengthwise in the first direction;

forming a dielectric feature over the substrate to fill space between two adjacent cladding layers;

forming a dummy gate structure over channel regions of the first and second fin-shaped active regions, the dummy gate structure orientated lengthwise in a second direction substantially perpendicular to the first direction and covering a first portion of the cladding layers;

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after the forming of the dummy gate structure, performing an etching process to remove a second portion of the cladding layers not covered by the dummy gate structure, thereby forming sidewall spacer trenches; forming a dielectric spacer in each of the sidewall spacer trenches; and

after the forming of the dielectric spacers, recessing portions of the first and second fin-shaped active regions not covered by the dummy gate structure, thereby forming source/drain openings;

forming source/drain features in the source/drain openings.

11. The method of claim 10, wherein the forming of the dielectric spacers comprises:

depositing a conformal dielectric layer over the substrate; and

selectively recessing the conformal dielectric layer to form the dielectric spacer in each of the sidewall spacer trenches,

wherein the selectively recessing of the conformal dielectric layer further forms gate spacers extending along sidewalls of the dummy gate structure.

12. The method of claim 10, wherein the forming of the source/drain features comprises:

epitaxially forming a first semiconductor layer in the source/drain openings;

epitaxially forming a second semiconductor layer over the first semiconductor layer; and

epitaxially forming a third semiconductor layer over the second semiconductor layer to fill the source/drain openings,

wherein a dopant concentration of the third semiconductor layer is greater than a dopant concentration of the second semiconductor layer.

13. The method of claim 12, wherein the first semiconductor layer, the second semiconductor layer, and the third semiconductor layer are in direct contact with the dielectric spacer.

14. The method of claim 10, further comprising:

after the forming of the source/drain features, selectively removing the dummy gate structure to form a gate trench, the gate trench exposing the first portion of the cladding layers;

selectively removing the first portion of the cladding layers; and

forming a gate stack over the substrate to fill the gate trench.

15. The method of claim 10, wherein the performing of the etching process selectively removes the second portion of the cladding layers without substantially etching the first

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and second fin-shaped active regions, and a height of the dielectric spacer is substantially equal to a height of the first and second fin-shaped active regions.

16. The method of claim 10, wherein a top surface of the dielectric spacer is below top surfaces of the source/drain features.

17. A method, comprising:

forming a fin-shaped structure over a substrate having a first region and a second region;

forming a cladding layer extending along a sidewall surface of the fin-shaped structure, the cladding layer having a first portion adjacent to the first region and a second portion adjacent to the second region;

forming a dielectric fin adjacent to the cladding layer, wherein when viewed from top, the fin-shaped structure, the cladding layer and the dielectric fin extend lengthwise along a same first direction;

forming a dummy gate structure covering the first portion of the cladding layer;

after the forming of the dummy gate structure, replacing the second portion of the cladding layer with a dielectric layer;

after forming the dielectric layer, recessing the second region of the fin-shaped structure to form a source/drain opening; and

forming a source/drain feature in the source/drain opening.

18. The method of claim 17, further comprising:

forming a gate spacer extending along a sidewall surface of the dummy gate structure.

19. The method of claim 17, wherein the fin-shaped structure comprises a plurality of channel layers interleaved by a plurality of sacrificial layers, a composition of the plurality of channel layers is different than a composition of the plurality of sacrificial layers, and the plurality of sacrificial layers and the cladding layer have the same composition, and the method further comprises:

performing an etching process to selectively remove the dummy gate structure, the plurality of sacrificial layers and the first portion of the cladding layer; and

forming a gate stack to wrap around the plurality of channel layers.

20. The method of claim 17, wherein the second portion of the cladding layer is disposed laterally adjacent to the first portion of the cladding layer along the first direction, and the dummy gate structure is in direct contact with both the second portion of the cladding layer and the fin-shaped structure.

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